COURSE:

CIP-B106 MOBILE & IOT FORENSICS

Case Study 3 - Echo Show 8 IoT Evidence Investigation

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1. Executive Summary

I examined the Echo Show 8 training eMMC image (EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin, 7,818,182,656 bytes) on a Kali Linux virtual machine. The image was attached as a read-only loop device and its ext4 partitions were mounted read-only. The disk uses a GPT table with 23 partitions, and the system (p21) and userdata (p23) partitions were readable without encryption.

The device identifies as an Amazon AEOAT ("athena") running Fire OS 7.5.6.0 on Android 9. Its Bluetooth name is "Echo Show 8 (2nd Gen)-3SD", its serial is G6G1GG10141403SD and its friendly name is "Amazing Assistant". There is one Android user (ID 0, "Sarfraz"), one Amazon account and three Prime Video profiles.

The userdata partition holds two saved Wi-Fi networks, a paired MacBook and Bluetooth speaker, YouTube, Google and Bing activity in the built-in web view, a Microsoft account linked for calendar access, five Prime Video titles, Amazon Music items shown on the home screen, and six photos and two videos taken with Photo Booth and uploaded.

2. Evidence and Method

2.1 Evidence Register

Item	Detail
Evidence file	EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin (file date 29 Dec 2023)
Location	VirtualBox shared folder /media/sf_EMMC_8GTF4R_ROM1_000000000000_0001D2000000
Size	7,818,182,656 bytes (15,269,888 sectors of 512 bytes)
MD5	b58fa04967161c158723c7b00a636533
SHA-256	8258ec28af667c07b429c5509a0b0e3b22e59da4331d6133528bfada9733c1d2
Userdata image	userdata_partition.bin, 3,758,079,488 bytes. I extracted it with dd from sectors 7,929,856 to 15,269,854 (7,339,999 sectors) of the eMMC image. It is a derived copy, not a separately supplied item.
Working copies	Databases and log files copied to /home/kali/case/ and /tmp/ before opening where SQLite or editing needed write access
Platform	Kali Linux VM (VirtualBox), 2 CPU threads
Examination dates	13 to 18 October 2026

I computed MD5 and SHA-256 before any other step and used them as the reference values (Figure 1).

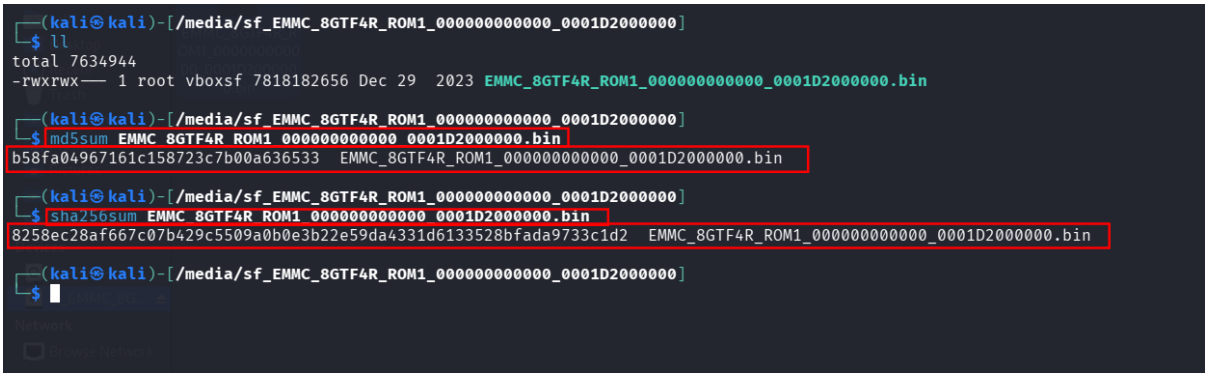


Figure 1: File listing and MD5/SHA-256 hashes of the eMMC image

2.2 Tools

Tool	Version	Purpose
losetup / mount (util-linux)	Kali package	Read-only loop device with partition scan; ext4 mounted ro,noload
The Sleuth Kit mmls	Kali package	GPT partition layout
sqlite3	3.46.1	Queries on Android and Amazon databases
jq	Kali package	Parsing JSON logs and cache files
bulk_extractor	2.1.1	Feature extraction (email, URL, MAC, JSON) over the full image
scalpel	1.60	Header carving of JPG and XML from the userdata partition
foremost	1.5.7	JPG and MP4 carving over the full image
PhotoRec	7.2	Carving from userdata unallocated space
ExifTool	13.55	Metadata of carved photos
dd, strings, grep, date, ImageMagick display	Kali packages	Extraction, string search, time conversion, image viewing

2.3 Mounting the Image

I attached the image with losetup --partscan --find --show --read-only, which gave /dev/loop0 and 23 partition devices (Figures 2 and 3). mmls confirmed a GUID Partition Table and the start sector of each partition (Figure 4).

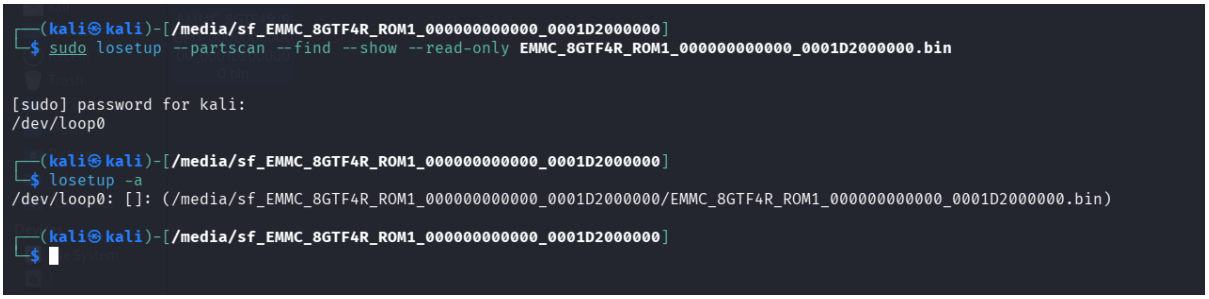


Figure 2: Image attached read-only as /dev/loop0

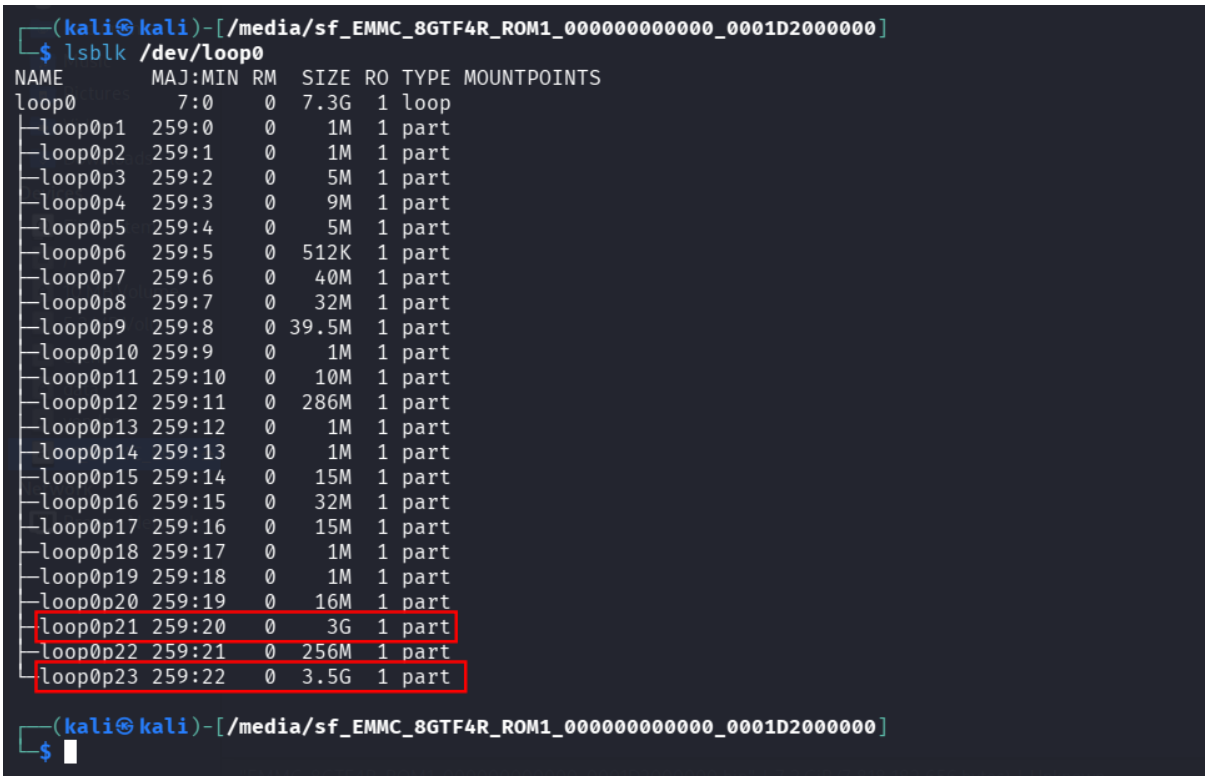


Figure 3: lsblk showing 23 partitions; p21 (3 GB) and p23 (3.5 GB) highlighted

```
┌──(kali㉿kali)-[/media/sf_EMMC_8GTF4R_ROM1_000000000000_0001D2000000]
└─$ mmls EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin
GUID Partition Table (EFI)
Offset Sector: 0
Units are in 512-byte sectors

     Slot      Start        End          Length       Description
000:  Meta      0000000000   0000000000   0000000001   Safety Table
001:  ————————  0000000000   0000000063   0000000064   Unallocated
002:  Meta      0000000001   0000000001   0000000001   GPT Header
003:  Meta      0000000002   0000000033   0000000032   Partition Table
004:  000       0000000064   0000002111   0000002048   kb
005:  001       0000002112   0000004159   0000002048   dkb
006:  002       0000004160   0000014399   0000010240   tee1
007:  003       0000014400   0000032767   0000018368   tee2
008:  004       0000032768   0000043007   0000010240   persist
009:  005       0000043008   0000044031   0000001024   misc
010:  006       0000044032   0000125951   0000081920   recovery
011:  007       0000125952   0000191487   0000065536   swdl
012:  008       0000191488   0000272383   0000080896   metadata
013:  009       0000272384   0000274431   0000002048   boot_para
014:  010       0000274432   0000294911   0000020480   nvcfg
015:  011       0000294912   0000880639   0000585728   vendor
016:  012       0000880640   0000882687   0000002048   spmfw
017:  013       0000882688   0000884735   0000002048   sspm_1
018:  014       0000884736   0000915455   0000030720   cam_vpu1
019:  015       0000915456   0000980991   0000065536   cam_vpu2
020:  016       0000980992   0001011711   0000030720   cam_vpu3
021:  017       0001011712   0001013759   0000002048   lk
022:  018       0001013760   0001015807   0000002048   lk2
023:  019       0001015808   0001048575   0000032768   boot
024:  020       0001048576   0007405567   0006356992   system
025:  021       0007405568   0007929855   0000524288   cache
026:  022       0007929856   0015269854   0007339999   userdata
027:  ————————  0015269855   0015269887   0000000033   Unallocated

┌──(kali㉿kali)-[/media/sf_EMMC_8GTF4R_ROM1_000000000000_0001D2000000]
└─$
```

Figure 4: mmls partition layout of the eMMC image

The system partition (p21) mounted at /mnt/echoshow8, and the kernel reported the source as write-protected and mounted it read-only (Figure 5). The userdata partition (p23) refused a plain read-only mount because ext4 wanted to replay its journal, which needs write access (dmesg, Figure 6). Mounting with -o ro,noload skips the replay and leaves the image untouched (Figure 7).

```
┌──(kali㉿kali)-[/media/sf_EMMC_8GTF4R_ROM1_000000000000_0001D2000000]
└─$ sudo mkdir /mnt/echoshow8

┌──(kali㉿kali)-[/media/sf_EMMC_8GTF4R_ROM1_000000000000_0001D2000000]
└─$ sudo mount /dev/loop0p21 /mnt/echoshow8
mount: /mnt/echoshow8: WARNING: source write-protected, mounted read-only.

┌──(kali㉿kali)-[/media/sf_EMMC_8GTF4R_ROM1_000000000000_0001D2000000]
└─$ cd /mnt/echoshow8

┌──(kali㉿kali)-[/mnt/echoshow8]
└─$ ls
acct        charger  default.prop  init.aosp.rc      init.inlocservice.rc   init.rc                 init.whad_cc.rc  odm      product  sys          verity_key
bin         config   dev           init.build.rc     init.meshmgrservice.rc init.recovery.mt8183.rc init.zygote32.rc  oem      sbin     system
bugreports  d        etc           init.environ.rc   init.perfmonitord.rc   init.usb.configfs.rc    lost+found       persist  sdcard   ueventd.rc
cache       data     init          init.fosflags.sh  init.perfrecoveryd.rc  init.usb.rc             mnt              proc     storage  vendor

┌──(kali㉿kali)-[/mnt/echoshow8]
└─$
```

Figure 5: System partition (p21) mounted read-only at /mnt/echoshow8

```
┌──(kali㉿kali)-[/mnt/echoshow8]
└─$ sudo mkdir /mnt/echoshow8-2

┌──(kali㉿kali)-[/mnt/echoshow8]
└─$ sudo mount /dev/loop0p23 /mnt/echoshow8-2
mount: /mnt/echoshow8-2: cannot mount /dev/loop0p23 read-only.
       dmesg(1) may have more information after failed mount system call.

┌──(kali㉿kali)-[/mnt/echoshow8]
└─$ sudo dmesg | tail -n 1
[ 9410.724816] EXT4-fs (loop0p23): write access unavailable, cannot proceed (try mounting with noload)

┌──(kali㉿kali)-[/mnt/echoshow8]
└─$
```

Figure 6: Userdata mount refused; dmesg asks for noload

```
┌──(kali㉿kali)-[/mnt/echoshow8]
└─$ sudo mount -o ro,noload /dev/loop0p23 /mnt/echoshow8-2

┌──(kali㉿kali)-[/mnt/echoshow8]
└─$ cd /mnt/echoshow8-2

┌──(kali㉿kali)-[/mnt/echoshow8-2]
└─$ ls
adb  app-asec       app-private  bootchart     data  local       media      misc     nfc          property              ss          system_de  user_de     vendor_de
anr  app-ephemeral  autotrace    cache         drm   logd        mediadrm   misc_ce  ota          resource-cache        system      tombstones vendor      vitals
app  app-lib        backup       dalvik-cache  fht   lost+found  metrics    misc_de  ota_package  securedStorageLocation system_ce user       vendor_ce  vp

┌──(kali㉿kali)-[/mnt/echoshow8-2]
└─$
```

Figure 7: Userdata partition (p23) mounted with ro,noload at /mnt/echoshow8-2

For the carving tools I also cut the userdata partition out as its own file using the start sector and length from mmls (Figures 8 and 9). The output size, 3,758,079,488 bytes, equals 7,339,999 × 512.

```
┌──(kali㉿kali)-[/mnt/echoshow8]
└─$ cd /media/sf_EMMC_8GTF4R_ROM1_000000000000_0001D2000000
mmls EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin
GUID Partition Table (EFI)
Offset Sector: 0
Units are in 512-byte sectors

     Slot    Start        End          Length       Description
000:  Meta    0000000000   0000000000   0000000001   Safety Table
001:  ------  0000000000   0000000063   0000000064   Unallocated
002:  Meta    0000000001   0000000001   0000000001   GPT Header
003:  Meta    0000000002   0000000033   0000000032   Partition Table
004:  000     0000000064   0000002111   0000002048   kb
005:  001     0000002112   0000004159   0000002048   dkb
006:  002     0000004160   0000014399   0000010240   tee1
007:  003     0000014400   0000032767   0000018368   tee2
008:  004     0000032768   0000043007   0000010240   persist
009:  005     0000043008   0000044031   0000001024   misc
010:  006     0000044032   0000125951   0000081920   recovery
011:  007     0000125952   0000191487   0000065536   swdl
012:  008     0000191488   0000272383   0000080896   metadata
013:  009     0000272384   0000274431   0000002048   boot_para
014:  010     0000274432   0000294911   0000020480   nvcfg
015:  011     0000294912   0000880639   0000585728   vendor
016:  012     0000880640   0000882687   0000002048   spmfw
017:  013     0000882688   0000884735   0000002048   sspm_1
018:  014     0000884736   0000915455   0000030720   cam_vpu1
019:  015     0000915456   0000980991   0000065536   cam_vpu2
020:  016     0000980992   0001011711   0000030720   cam_vpu3
021:  017     0001011712   0001013759   0000002048   lk
022:  018     0001013760   0001015807   0000002048   lk2
023:  019     0001015808   0001048575   0000032768   boot
024:  020     0001048576   0007405567   0006356992   system
025:  021     0007405568   0007929855   0000524288   cache
026:  022     0007929856   0015269854   0007339999   userdata
027:  ------  0015269855   0015269887   0000000033   Unallocated

┌──(kali㉿kali)-[/media/sf_EMMC_8GTF4R_ROM1_000000000000_0001D2000000]
└─$
```

Figure 8: Userdata start sector (7,929,856) and length (7,339,999 sectors) taken from mmls

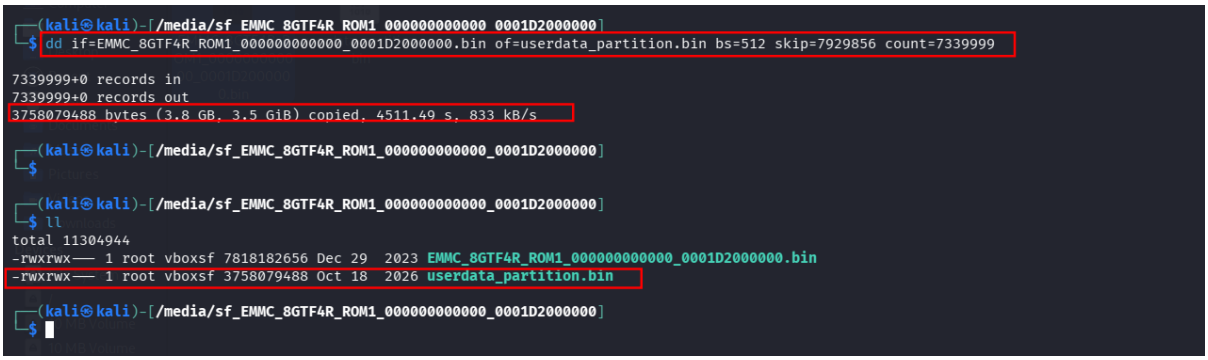


Figure 9: dd extraction of the userdata partition to userdata_partition.bin

Handling note. userdata_partition.bin and the scalpel output were written into the shared folder next to the evidence file. This did not change the .bin file, but in casework I would keep derived files on separate storage and re-hash the original at the end.

2.4 Bulk Extraction

bulk_extractor 2.1.1 ran over the whole image with output in ~/bulkext_raw (Figure 10). It read all 7,818,182,656 bytes and produced feature files and histograms for email addresses, URLs, MAC addresses (ether), JSON and searches (Figures 11 and 12).

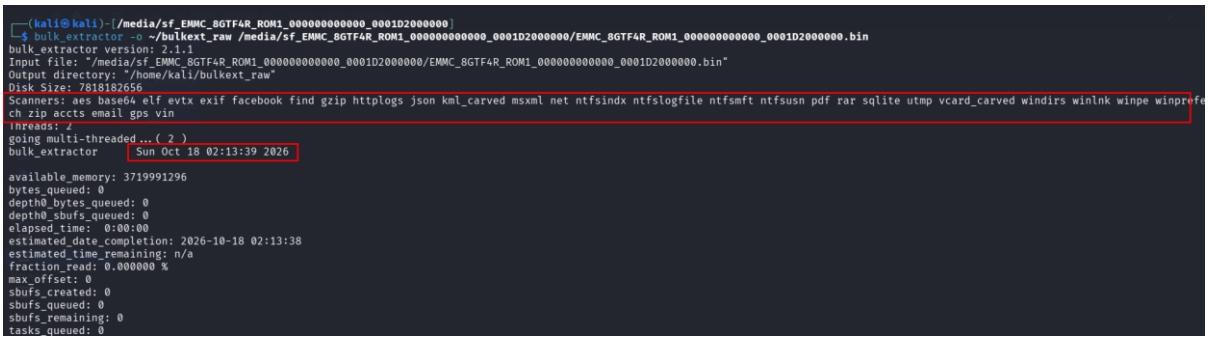


Figure 10: bulk_extractor run against the eMMC image

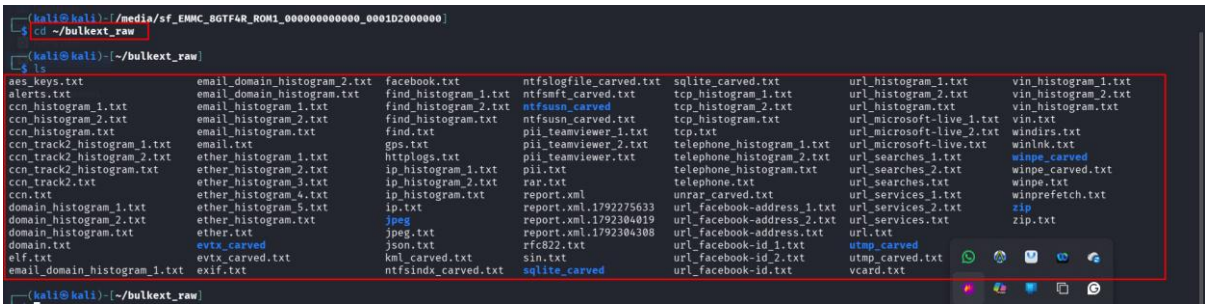


Figure 11: bulk_extractor output files

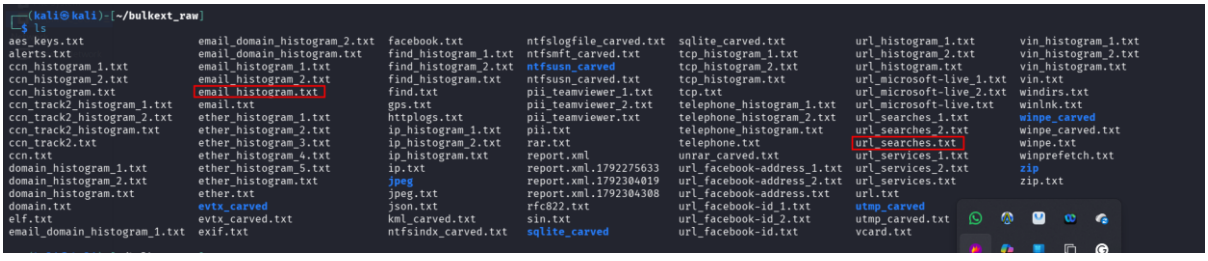


Figure 12: email_histogram.txt and url_searches.txt in the bulk_extractor output

2.5 Timestamp Conversion

Format	Where seen	Conversion
Unix epoch, milliseconds	Android XML, accounts_de, Prime Video, Zordon	Divide by 1000, then date -u -d @<seconds>
Unix epoch, seconds	build.prop, bt_config.conf	date -u -d @<seconds>
Chromium/WebKit time (µs since 1601-01-01)	Web view History database	(value ÷ 1,000,000) − 11,644,473,600, then date -u
ISO 8601 with offset	alexa-anchor log	Offset is −04:00; add 4 hours for UTC
Local time in a filename or text	Photo Booth filenames, accounts_de debug_table	Compared against a UTC value for the same event

All converted times in this report are UTC unless stated. The device was set to US Eastern Daylight Time (UTC−4) for most of the activity: the alexa-anchor records carry −0400, and each Photo Booth filename sits about four hours behind its upload record. Entries written during initial setup on 2 October 2023 match UTC exactly, which suggests the time zone was set during setup (see Section 3.4).

3. Device and User Profile

3.1 Partition Layout

The GPT holds 23 named partitions. Most are boot and firmware areas (kb, dkb, tee1, tee2, lk, lk2, boot, recovery, cam_vpu1 to cam_vpu3) with no mountable filesystem. The ext4 partitions are p5 (persist), p11 (nvcfg), p12 (vendor), p21 (system, start 1,048,576, 6,356,992 sectors), p22 (cache) and p23 (userdata, start 7,929,856, 7,339,999 sectors). I repeated mmls while profiling the device (Figure 13).

```
┌──(kali㉿kali)-[/media/sf_EMMC_8GTF4R_ROM1_000000000000_0001D2000000]
└─$ mmls EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin
GUID Partition Table (EFI)
Offset Sector: 0
Units are in 512-byte sectors

     Slot      Start        End          Length       Description
000:  Meta     0000000000   0000000000   0000000001   Safety Table
001:  ------   0000000000   0000000063   0000000064   Unallocated
002:  Meta     0000000001   0000000001   0000000001   GPT Header
003:  Meta     0000000002   0000000033   0000000032   Partition Table
004:  000      0000000064   0000002111   0000002048   kb
005:  001      0000002112   0000004159   0000002048   dkb
006:  002      0000004160   0000014399   0000010240   tee1
007:  003      0000014400   0000032767   0000018368   tee2
008:  004      0000032768   0000043007   0000010240   persist
009:  005      0000043008   0000044031   0000001024   misc
010:  006      0000044032   0000125951   0000081920   recovery
011:  007      0000125952   0000191487   0000065536   swdl
012:  008      0000191488   0000272383   0000080896   metadata
013:  009      0000272384   0000274431   0000002048   boot_para
014:  010      0000274432   0000294911   0000020480   nvcfg
015:  011      0000294912   0000880639   0000585728   vendor
016:  012      0000880640   0000882687   0000002048   spmfw
017:  013      0000882688   0000884735   0000002048   sspm_1
018:  014      0000884736   0000915455   0000030720   cam_vpu1
019:  015      0000915456   0000980991   0000065536   cam_vpu2
020:  016      0000980992   0001011711   0000030720   cam_vpu3
021:  017      0001011712   0001013759   0000002048   lk
022:  018      0001013760   0001015807   0000002048   lk2
023:  019      0001015808   0001048575   0000032768   boot
024:  020      0001048576   0007405567   0006356992   system
025:  021      0007405568   0007929855   0000524288   cache
026:  022      0007929856   0015269854   0007339999   userdata
027:  ------   0015269855   0015269887   0000000033   Unallocated

┌──(kali㉿kali)-[/media/sf_EMMC_8GTF4R_ROM1_000000000000_0001D2000000]
└─$
```

Figure 13: mmls output repeated during device profiling; system and userdata highlighted

3.2 Operating System and Build

Source: system partition, /system/build.prop (Figures 14 and 15).

Property	Value
ro.product.brand / model	Amazon / AEOAT
ro.product.name / device	athena / athena
ro.build.version.release (SDK)	9 (28)
ro.build.display.id	PS7560.3677N
ro.build.version.security_patch	2022-01-01
ro.build.date.utc	1695264110 = 2023-09-21 02:41:50 UTC

Property	Value
ro.build.mktg.fireos / version.name	Fire OS 7.5.6.0 (PS7560/3677)
ro.product.package_name	com.amazon.athena.android.os
ro.build.lab126.project	athena_fireos_ship_7560

AEOAT is an internal model code, not a product name. The link to "Echo Show 8 (2nd Gen)" comes from the Bluetooth adapter name in bt_config.conf (Section 4.1), not from the course slides.

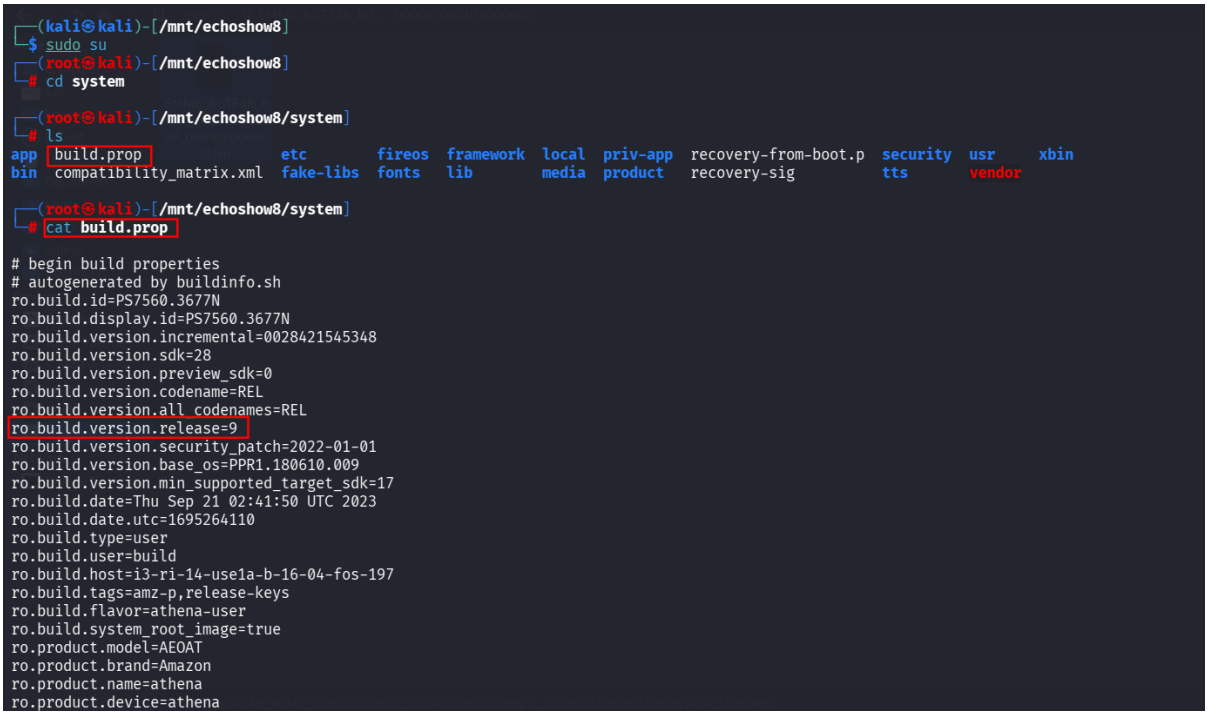


Figure 14: build.prop: build ID, Android release 9, build date and model AEOAT

```
# maximum backup size for factory reset whitelist in kB
ro.recovery.wl.maxstore=921600

# Reserve WHA server ports.
sys.sysctl.ip_loc_res_ports=55442-55445

# Allow boosting of CPUs up to fifteen seconds.
persist.perfboost.time=15000

# Extended smarthome discovery timeout in msec
persist.sh_discovery_timeout=40000

#enable OTA download via metered network
persist.sys.ota_allow_metered=true
#
# ADDITIONAL_BUILD_PROPERTIES
#
ro.treble.enabled=true
persist.sys.dalvik.vm.lib.2=libart.so
dalvik.vm.isa.arm.variant=cortex-a53
dalvik.vm.isa.arm.features=default
net.bt.name=Android
dalvik.vm.stack-trace-dir=/data/anr
ro.build.lab126.buildtype=Nightly
ro.build.lab126.project=athena_fireos_ship_7560
ro.build.lab126.build=3677
ro.build.version.serial=3677
ro.build.configuration=knightfamily
ro.build.version.fireos=7.0
ro.build.version.number=0028421545348
ro.build.mktg.fireos=Fire OS 7.5.6.0
ro.build.version.name=Fire OS 7.5.6.0 (PS7560/3677)
ro.product.package_name=com.amazon.athena.android.os
ro.expect.recovery_id=0×eb4b1c0196bad041b21785b5f1087dedb9c2d73c0000000000000000000000000


┌──(root㉿kali)-[/mnt/echoshow8/system]
└─#
```

Figure 15: build.prop: Fire OS 7.5.6.0 and package name

/system/app lists the preinstalled applications, including Bluetooth, Calendar, Email, Music, AmazonWebView and KnightCaptivePortalLogin (Figure 16). These show what the device can do, not what was used.

```
┌──(root㉿kali)-[/mnt/echoshow8/system]
└─# ls
app  build.prop              etc        fireos  framework  local  priv-app  recovery-from-boot.p  security  usr     xbin
bin  compatibility_matrix.xml  fake-libs  fonts   lib        media  product   recovery-sig          tts       vendor

┌──(root㉿kali)-[/mnt/echoshow8/system]
└─# cd app

┌──(root㉿kali)-[/mnt/echoshow8/system/app]
└─# ls
AmazonDeviceMetrics          BookmarkProvider       CtsShimPrebuilt  HTMLViewer                 Music                      Protips                Traceur
AmazonWebView                BuiltInPrintService    DeskClock        KeyChain                   NfcNci                     QuickSearchBox         WallpaperBackup
AmazonWirelessMetricsService  Calendar              Email            KnightBluetoothAVRCPClient  PacProcessor               SecureElement          WallpaperPicker
Bluetooth                    CertInstaller          ExtShared        KnightCaptivePortalLogin   PicoTts                    SidewalkDeviceService  WAPPushManager
BluetoothMidiService         CompanionDeviceManager  fdrw             LiveWallpapersPicker       PrintRecommendationService  SimAppDialog           whad_app

┌──(root㉿kali)-[/mnt/echoshow8/system/app]
└─#
```

Figure 16: Preinstalled system applications in /system/app

3.3 Device Identifiers

Value	Source (userdata)
Friendly name "Amazing Assistant"	data/com.android.settings/shared_prefs/deviceNameSharedPref.xml, key deviceNameKey
Serial G6G1GG10141403SD	com.amazon.settings.DNSVR_STORE.xml, devicepreferenceslist entry G6G1GG10141403SD|A0565892JJWBS2358X2D
Serial G6G1GG10141403SD, device type A15996VY63BQ2D, friendly name "Amazing Assistant", manufacturer amazon	data/amazon.speech.sim/files/device_capabilities_api.amazonalexa.com

Two separate files give the same serial number and the same friendly name, so I treat both values as corroborated.

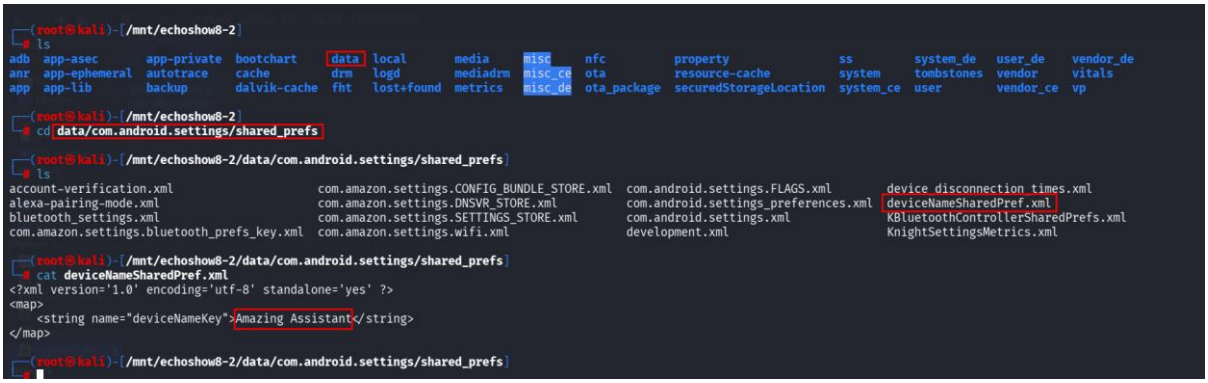


Figure 17: Device name "Amazing Assistant" in deviceNameSharedPref.xml

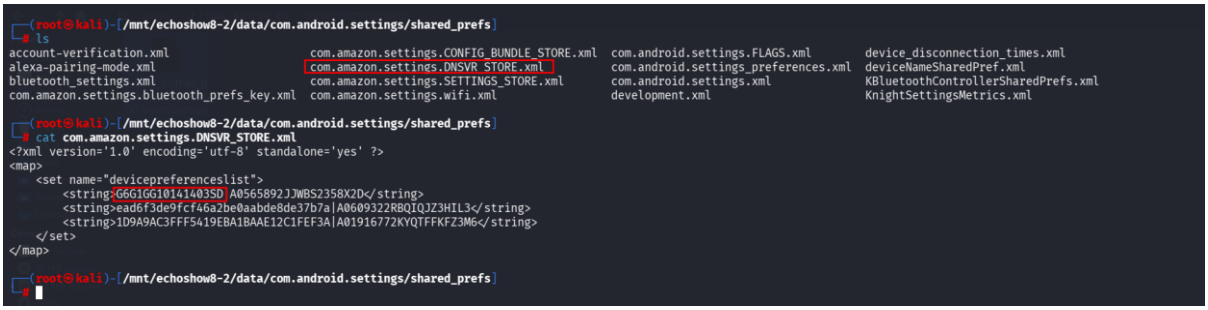


Figure 18: Serial number in com.amazon.settings.DNSVR_STORE.xml

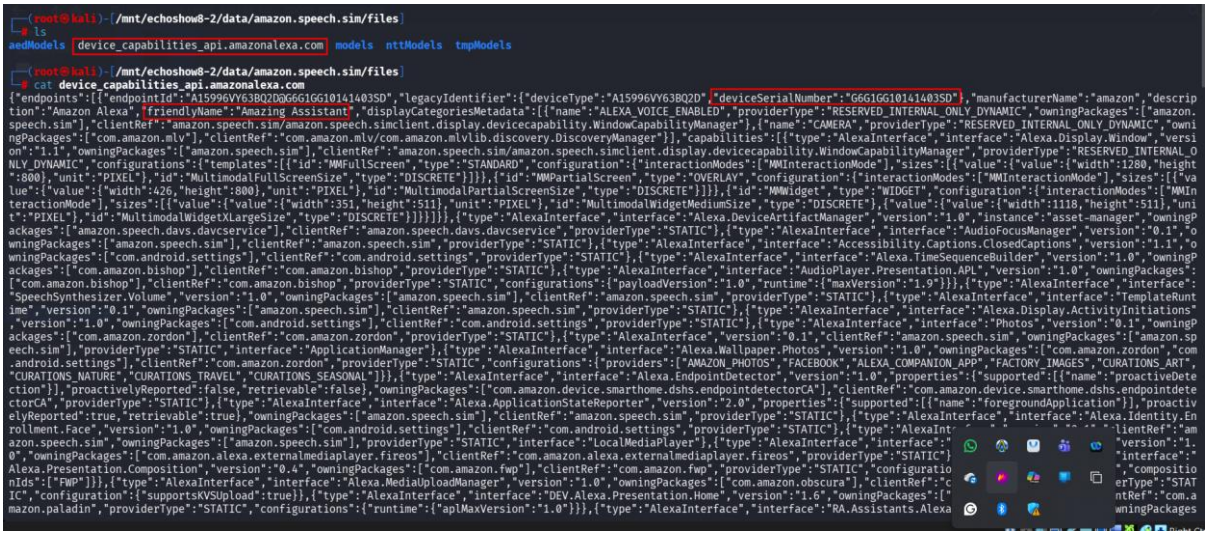


Figure 19: Serial number, device type and friendly name in the Alexa device capabilities file

3.4 Accounts and Users

Amazon account. data/com.amazon.imp/databases/map_data_storage_v2.db was copied to /tmp and queried. The accounts table holds amzn1.account.AFTJBUB4CIBBMH2S2AQDEY5QCN2A with the name "Sarfraz", and encryption_data holds a key_encryption_secret (Figures 20 and 21). The same account ID appears in the Photos (Zordon) database filename and in every Prime Video playback row, which ties those records to this account.

```
┌──(root㉿kali)-[/mnt/echoshow8-2/data]
└─# cd com.amazon.imp/databases

┌──(root㉿kali)-[/mnt/echoshow8-2/data/com.amazon.imp/databases]
└─# l
map_data_storage_v2.db
```

Figure 20: map_data_storage_v2.db in com.amazon.imp

```
┌──(root㉿kali)-[/mnt/echoshow8-2/data/com.amazon.imp/databases]
└─# cp map_data_storage_v2.db /tmp/
chown root:root /tmp/map_data_storage_v2.db
sqlite3 /tmp/map_data_storage_v2.db
SQLite version 3.46.1 2024-08-13 09:16:08
Enter ".help" for usage hints.
sqlite> .tables
account_data      android_metadata  encryption_data
accounts          device_data
sqlite> SELECT * FROM accounts;
2|amzn1.account.AFTJBUB4CIBBMH2S2AQDEY5QCN2A|Sarfraz
sqlite> SELECT * FROM encryption_data;
1|key_encryption_secret|LRxaKMwe0HJCljLR9XNdQw==

sqlite>
```

Figure 21: Amazon account ID and name "Sarfraz"; key encryption secret

Android users. system/users/userlist.xml and sorteduserlist.xml list one user, ID 0, with the default guest restrictions (no_sms, no_outgoing_calls, no_config_wifi, no_install_unknown_sources). nextSerialNumber is 10, so other user IDs were allocated at some point; I found nothing showing what they were. 0.xml names user 0 "Sarfraz", with lastLoggedIn 1698016596033 = 2023-10-22 23:16:36 UTC. Its lastLoggedInFingerprint contains PS7560.3677N, which matches build.prop (Figures 22 to 25).

```
┌──(kali㉿kali)-[/mnt/echoshow8-2]
└─$ ls
adb  app-asec       app-private  bootchart     data  local       media      misc     nfc          property              ss          system_de  user_de     vendor_de
anr  app-ephemeral  autotrace    cache         drm   logd        mediadrm   misc_ce  ota          resource-cache        system      tombstones vendor      vitals
app  app-lib        backup       dalvik-cache  fht   lost+found  metrics    misc_de  ota_package  securedStorageLocation system_ce user       vendor_ce  vp

┌──(kali㉿kali)-[/mnt/echoshow8-2]
└─$ cd system/users

┌──(kali㉿kali)-[/mnt/echoshow8-2/system/users]
└─$ ls
0  0.xml  sorteduserlist.xml  userlist.xml

┌──(kali㉿kali)-[/mnt/echoshow8-2/system/users]
└─$
```

Figure 22: system/users folder

```
┌──(kali㉿kali)-[/mnt/echoshow8-2/system/users]
└─$ cat userlist.xml
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<users nextSerialNumber="10" version="7">
    <guestRestrictions>
        <restrictions no_sms="true" no_install_unknown_sources="true" no_config_wifi="true" no_outgoing_calls="true" />
    </guestRestrictions>
    <deviceOwnerUserId id="-10000" />
    <user id="0" />
</users>

┌──(kali㉿kali)-[/mnt/echoshow8-2/system/users]
└─$
```

Figure 23: userlist.xml: single user ID 0 and guest restrictions

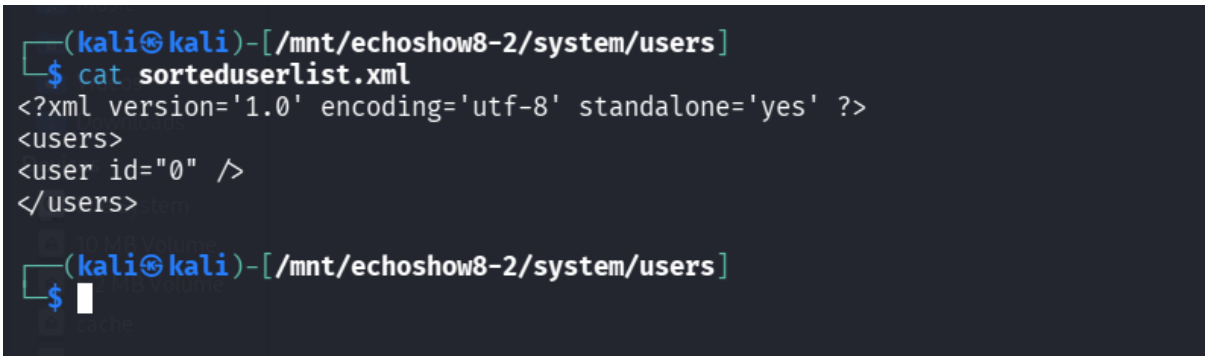


Figure 24: sorteduserlist.xml



Figure 25: 0.xml: user name Sarfraz and lastLoggedIn converted to UTC

Android accounts. system_ce/0/accounts_ce.db and system_de/0/accounts_de.db list four accounts: two AmazonSyncAccount entries, "Amazon Video Sync" and "Sarfraz" (type com.amazon.account). The meta table maps com.amazon.account to UID 10111. The debug_table records account changes (Figures 26 to 29):

debug_table entry	Time as logged	UID
action_account_add	2021-02-18 03:11:18	10118
action_account_add (x3)	2023-10-02 21:00:18, 21:00:26, 21:00:32	10111, 10029, 10118
action_called_account_remove / action_account_remove	2023-10-09 16:34:24	10111
action_account_add	2023-10-09 16:39:41	10111

Cross-source check. accounts_de stores the Sarfraz account time as 1696883981793 = 2023-10-09 20:39:41 UTC. The debug_table logs the re-add at 16:39:41, the same second four hours earlier, so that entry is in local EDT. Twenty-seven seconds later, the setup app cached an Amazon address response dated 20:40:08 GMT (Section 4.3). The account was removed and re-added on 9 October and the setup flow then ran again. The 2 October debug entries match the accounts_de values for accounts 3 and 4 (2023-10-02 21:00:26 and 21:00:32 UTC) with no offset, so the clock was still on UTC at that point. The first Photo Booth photo, about 50 minutes later, is already named in EDT.



Figure 26: system_ce/0 contents

```
┌──(kali㉿kali)-[/mnt/echoshow8-2/system_ce/0]
└─$ sqlite3 accounts_ce.db
SQLite version 3.46.1 2024-08-13 09:16:08
Enter ".help" for usage hints.
sqlite> .tables
accounts          android_metadata  authtokens        extras
sqlite> SELECT * FROM accounts;
1|AmazonSyncAccount-cba1f8294651551b7d9d16bef0970fdd79fca92f414c3338137185f6acf40f71|amazon.account|
3|Amazon Video Sync|amazon.account|
4|AmazonSyncAccount-6957b355b974dde07bb47bf8cd86218da71f3ce800784526ba4345cda0abf63b|amazon.account|
5|Sarfraz|com.amazon.account|
sqlite>
```

Figure 27: accounts_ce.db accounts table

```
┌──(kali㉿kali)-[/mnt/echoshow8-2]
└─$ ls
adb  app-asec       app-private  bootchart     data  local       media     misc     nfc          property              ss          system_de  user_de     vendor_de
anr  app-ephemeral  autotrace    cache         drm   logd        mediadrm  misc_ce  ota          resource-cache        system      tombstones vendor      vitals
app  app-lib        backup       dalvik-cache  fht   lost+found  metrics   misc_de  ota_package  securedStorageLocation  system_ce  user       vendor_ce   vp

┌──(kali㉿kali)-[/mnt/echoshow8-2]
└─$ cd system_de/0

┌──(kali㉿kali)-[/mnt/echoshow8-2/system_de/0]
└─$ ls
accounts_de.db  accounts_de.db-journal  persisted_taskIds.txt  ringtones

┌──(kali㉿kali)-[/mnt/echoshow8-2/system_de/0]
└─$
```

Figure 28: system_de/0 contents

```
┌──(kali㉿kali)-[/mnt/echoshow8-2/system_de/0]
└─$ sqlite3 accounts_de.db
SQLite version 3.46.1 2024-08-13 09:16:08
Enter ".help" for usage hints.
sqlite> .tables
accounts          debug_table       meta              visibility
android_metadata  grants            shared_accounts
sqlite> SELECT * FROM accounts;
1|AmazonSyncAccount-cba1f8294651551b7d9d16bef0970fdd79fca92f414c3338137185f6acf40f71|amazon.account||1613617878925
3|Amazon Video Sync|amazon.account||1696280426425
4|AmazonSyncAccount-6957b355b974dde07bb47bf8cd86218da71f3ce800784526ba4345cda0abf63b|amazon.account||1696280432902
5|Sarfraz|com.amazon.account||1696883981793
sqlite> SELECT * FROM meta;
auth_uid_for_type:com.amazon.account|10111
auth_uid_for_type:com.android.email|10126
auth_uid_for_type:com.android.exchange|10126
auth_uid_for_type:amazon.account|10118
sqlite> SELECT * FROM debug_table;
1|action_account_add|2021-02-18 03:11:18|10118|accounts|0
2|action_account_add|2023-10-02 21:00:18|10111|accounts|1
3|action_account_add|2023-10-02 21:00:26|10029|accounts|2
4|action_account_add|2023-10-02 21:00:32|10118|accounts|3
2|action_called_account_remove|2023-10-09 16:34:24|10111|accounts|4
2|action_account_remove|2023-10-09 16:34:24|10111|accounts|5
5|action_account_add|2023-10-09 16:39:41|10111|accounts|6
sqlite>
```

Figure 29: accounts_de.db accounts, meta and debug_table

3.5 Usage Statistics

system/usagestats/0/daily holds nine daily files named by epoch milliseconds, from 1697329601115 (2023-10-15 00:26:41 UTC) to 1698022332424 (2023-10-23 00:52:12 UTC). Older days have been rolled off by Android, so usage before 15 October is not available from this source.

In the file starting 2023-10-22 00:45:12 UTC, com.amazon.paladin (the home screen) has appLaunchCount 5 and timeActive 7,898,142 ms (2 h 11 m). Most other packages have a negative lastTimeActive, meaning no foreground use that day. Event-log times are offsets from the start of the file, and the MMLauncherActivity entry is type 1 (moved to foreground) (Figures 30 to 32).

```
┌──(kali㉿kali)-[/mnt/echoshow8-2/system/usagestats/0]
└─$ ls
daily  monthly  version  weekly  yearly

┌──(kali㉿kali)-[/mnt/echoshow8-2/system/usagestats/0]
└─$ cd daily

┌──(kali㉿kali)-[/mnt/…/system/usagestats/0/daily]
└─$ ls
1697329601115  1697416099195  1697502596344  1697589171089  1697675612477  1697762397254  1697848962840  1697935512409  1698022332424

┌──(kali㉿kali)-[/mnt/…/system/usagestats/0/daily]
└─$ date -u -d @1697935512
Sun Oct 22 12:45:12 AM UTC 2023

┌──(kali㉿kali)-[/mnt/…/system/usagestats/0/daily]
└─$
```

Figure 30: Daily usage stats files; one filename converted to UTC

```
┌──(kali㉿kali)-[/mnt/…/system/usagestats/0/daily]
└─$ cat 1697935512409 | more
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<usagestats version="1" endTime="85500007">
    <interactive count="11" time="1697860050653" />
    <non-interactive count="11" time="1697859985758" />
    <keyguard-shown count="0" time="0" />
    <keyguard-hidden count="0" time="0" />
    <packages>
        <package lastTimeActive="-1697775240571" package="com.android.providers.calendar" timeActive="0" lastEvent="0" />
        <package lastTimeActive="78119740" package="com.amazon.device.rdmapplication" timeActive="0" lastEvent="0" />
        <package lastTimeActive="-1697775240571" package="com.amazon.whisperjoin.middleware" timeActive="0" lastEvent="0" />
        <package lastTimeActive="-1697775240571" package="com.amazon.alexa.beaconbroadcaster" timeActive="0" lastEvent="0" />
        <package lastTimeActive="84396503" package="com.amazon.alexa.datastore.app" timeActive="0" lastEvent="0" />
        <package lastTimeActive="59193390" package="com.amazon.alexa.ms.kinex" timeActive="0" lastEvent="0" />
        <package lastTimeActive="-1697775240571" package="com.amazon.device.smarthome.dshs.services" timeActive="0" lastEvent="0" />
        <package lastTimeActive="-1697775240571" package="com.amazon.imp" timeActive="0" lastEvent="0" />
        <package lastTimeActive="-1697775240571" package="com.android.launcher3" timeActive="0" lastEvent="0" />
        <package lastTimeActive="-1697775240571" package="com.amazon.tcomm" timeActive="0" lastEvent="0" />
        <package lastTimeActive="-1697775240571" package="com.amazon.NativeAccessorProxyServices" timeActive="0" lastEvent="0" />
        <package lastTimeActive="-1697775240571" package="amazon.speech.sim" timeActive="0" lastEvent="0" />
        <package lastTimeActive="-1697775240571" package="com.android.se" timeActive="0" lastEvent="0" />
        <package lastTimeActive="14374522" package="com.amazon.android.service.wifiprofilemanager" timeActive="0" lastEvent="0" />
        <package lastTimeActive="-1697775240571" package="android.ext.services" timeActive="0" lastEvent="0" />
        <package lastTimeActive="-1697775240571" package="com.amazon.knight.blink" timeActive="0" lastEvent="0" />
        <package lastTimeActive="-1697775240571" package="com.amazon.device.smarthome.adapters.echo" timeActive="0" lastEvent="0" />
        <package lastTimeActive="-1697775240571" package="com.amazon.device.smarthome.adapters.mock" timeActive="0" lastEvent="0" />
        <package lastTimeActive="-1697775240571" package="com.amazon.banyan.core" timeActive="0" lastEvent="0" />
        <package lastTimeActive="-1697775240571" package="com.amazon.device.software.ota" timeActive="0" lastEvent="0" />
        <package lastTimeActive="-1697775240571" package="amazon.speech.davs.davcservice" timeActive="0" lastEvent="0" />
        <package lastTimeActive="-1697775240571" package="com.amazon.knight.btavrcp" timeActive="0" lastEvent="0" />
        <package lastTimeActive="84810298" package="com.amazon.paladin" timeActive="7898142" lastEvent="2" appLaunchCount="5" />
        <package lastTimeActive="-1697775240571" package="com.amazon.alexa.timeoutmanagerapp" timeActive="0" lastEvent="0" />
```

Figure 31: Daily usage file: package list with paladin launch count and active time

```
        <package lastTimeActive="-1697775240571" package="com.amazon.device.gadgetscontrolmanager" timeActive="0" lastEvent="0" />
    </packages>
    <configurations />
    <event-log>
        <event time="3205245" package="com.amazon.alexa.datastore.app" flags="0" type="12" />
        <event time="3295445" package="com.amazon.android.service.wifiprofilemanager" flags="0" type="12" />
        <event time="7784509" package="com.amazon.alexa.datastore.app" flags="0" type="12" />
        <event time="10286912" package="com.amazon.alexa.datastore.app" flags="0" type="12" />
        <event time="10485765" package="com.amazon.alexa.datastore.app" flags="0" type="12" />
        <event time="12863303" package="com.amazon.device.rdmapplication" flags="0" type="12" />
        <event time="12864014" package="com.amazon.device.rdmapplication" flags="0" type="12" />
        <event time="14374522" package="com.amazon.android.service.wifiprofilemanager" flags="0" type="12" />
        <event time="17728673" package="com.amazon.alexa.datastore.app" flags="0" type="12" />
        <event time="17728917" package="com.amazon.alexa.datastore.app" flags="0" type="12" />
        <event time="18584708" package="com.amazon.alexa.datastore.app" flags="0" type="12" />
        <event time="23188181" package="com.amazon.alexa.datastore.app" flags="0" type="12" />
        <event time="24990376" package="com.amazon.alexa.datastore.app" flags="0" type="12" />
        <event time="26673439" package="android" flags="0" type="15" />
        <event time="26674042" package="com.amazon.paladin" class="com.amazon.paladin.app.MMLauncherActivity" flags="0" type="1" />
lore--
```

Figure 32: Daily usage file: event log

3.6 Recent Tasks and Snapshots

system_ce/0/recent_tasks holds task files 617, 623, 624 and 625, and system_ce/0/snapshots holds screen captures for 617, 623, 624, 625 and 627 (Figure 33). Snapshot 627 has no matching task file, which is consistent with the task being closed after the capture was saved.

- 617_task.xml: app com.amazon.cloud9 (SlateEchoActivity). Intent data http://awa/search?query=kapil+sharma+show&provider=youtube&locale=en-US. last_time_moved 1698022572750 = 2023-10-23 00:56:12 UTC (Figure 37).
- 617.jpg: a YouTube clip titled "Rakul Shares 'The Story Of Her Na...'", with a Sony channel logo, which fits the Kapil Sharma Show search in the task (Figure 35).
- 623.jpg: Bluetooth screen showing "Sarfraz's MacBook Pro" (Figure 36).
- 627.jpg: a reminder card "At 7:00 PM Exercise" (Figure 34).

A snapshot shows the last screen of a task. It does not show how long the screen was viewed or who was looking at it.



Figure 33: system_ce/0/snapshots folder

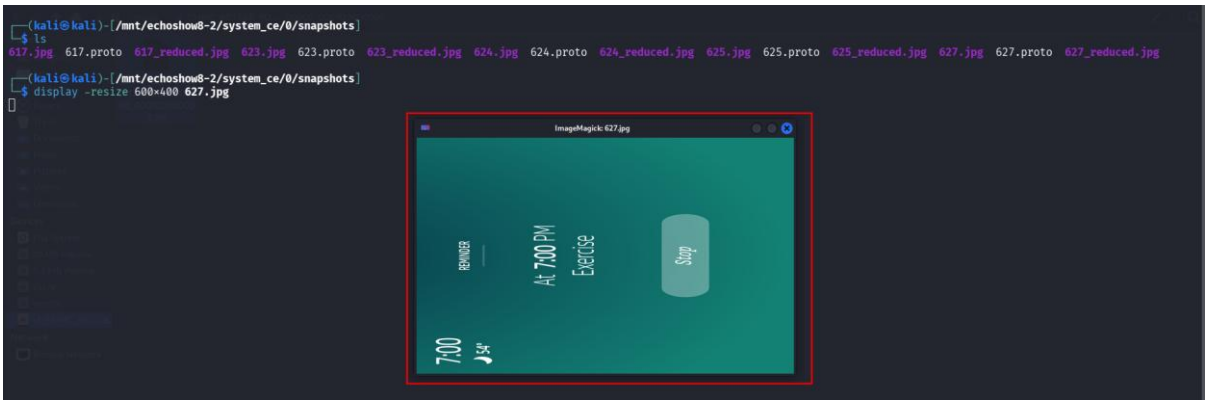


Figure 34: Snapshot 627.jpg: reminder "At 7:00 PM Exercise"

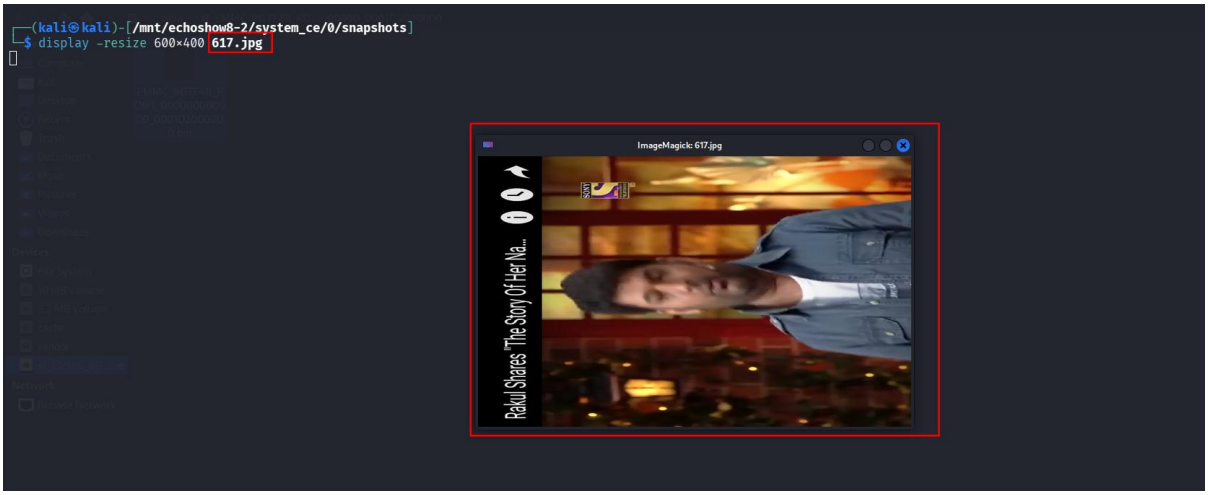


Figure 35: Snapshot 617.jpg: YouTube clip in the web view

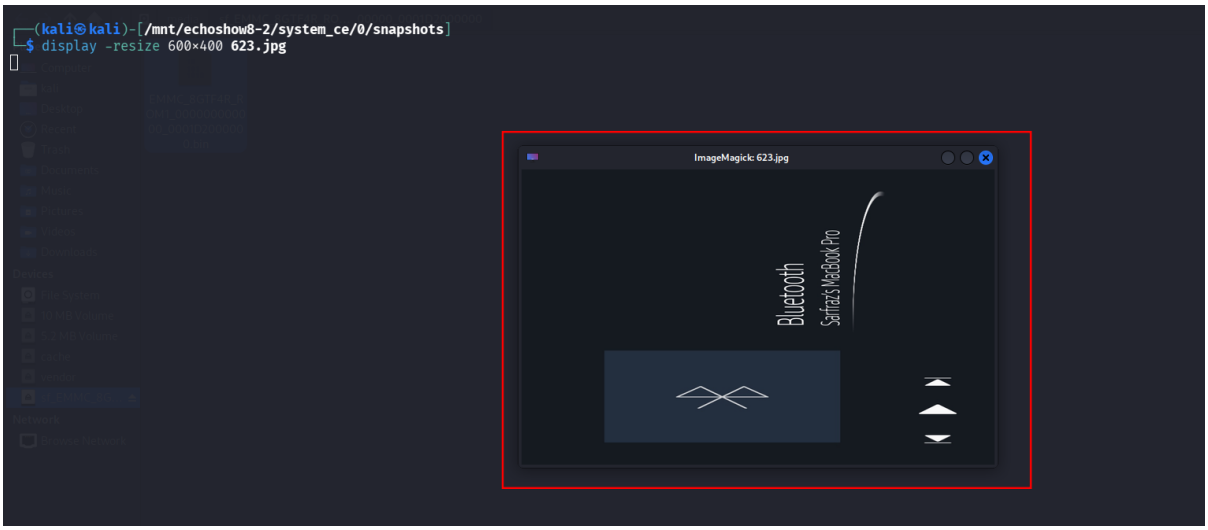


Figure 36: Snapshot 623.jpg: Bluetooth screen with Sarfraz's MacBook Pro

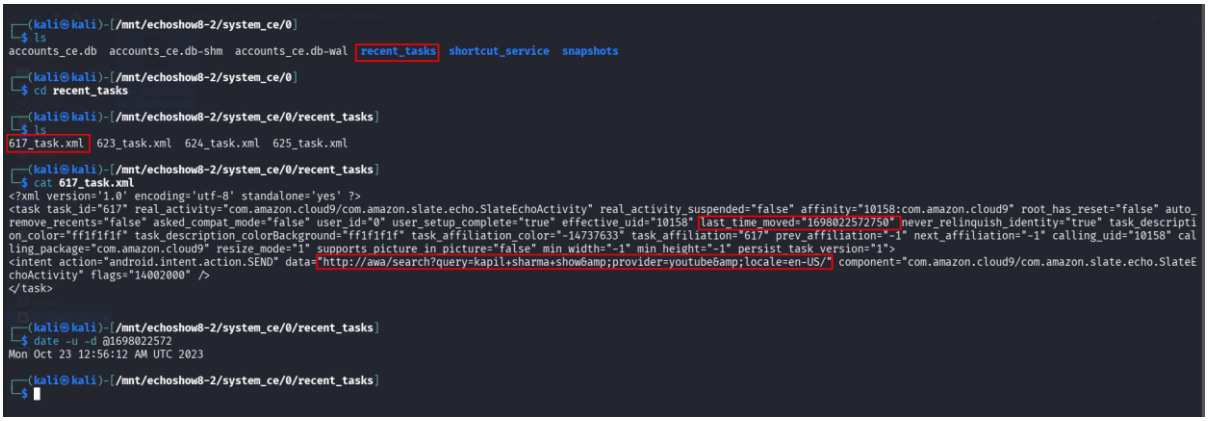


Figure 37: 617_task.xml: Kapil Sharma Show search intent and last_time_moved

4. Artefact Findings

4.1 Network Connectivity

Wi-Fi. misc/wifi/WifiConfigStore.xml holds two saved WPA-PSK networks, "Fios-pJK9q" and "TellMyWiFiLoveHer". Both use the pre-shared key cop576him662ill and both have ValidatedInternetAccess = true (Figures 38 and 39). This is stored configuration. It shows that each network was saved and reached the internet at some point, but it gives no connection times. The shared key suggests one router renamed or a reused password; the file cannot tell which.

```
┌──(root㉿kali)-[/mnt/echoshow8-2/misc/wifi]
└─# ls
sockets  softap.conf  WifiConfigStore.xml  wpa_supplicant

┌──(root㉿kali)-[/mnt/echoshow8-2/misc/wifi]
└─# cat WifiConfigStore.xml
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<WifiConfigStoreData>
<int name="Version" value="1" />
<NetworkList>
<Network>
<WifiConfiguration>
<string name="ConfigKey">&quot;Fios-pJK9q&quot;WPA_PSK</string>
<string name="SSID">&quot;Fios-pJK9q&quot;</string>
<null name="BSSID" />
<string name="PreSharedKey">&quot;cop576him662ill&quot;</string>
<int name="Priority" value="0" />
<null name="WEPKeys" />
<int name="WEPTxKeyIndex" value="0" />
<boolean name="HiddenSSID" value="false" />
<boolean name="RequirePMF" value="false" />
<byte-array name="AllowedKeyMgmt" num="1">02</byte-array>
<byte-array name="AllowedProtocols" num="1">03</byte-array>
<byte-array name="AllowedAuthAlgos" num="1">01</byte-array>
<byte-array name="AllowedGroupCiphers" num="1">0f</byte-array>
<byte-array name="AllowedPairwiseCiphers" num="1">06</byte-array>
<boolean name="Shared" value="true" />
<int name="Status" value="2" />
<null name="FQDN" />
<null name="ProviderFriendlyName" />
<null name="LinkedNetworksList" />
<null name="DefaultGwMacAddress" />
<boolean name="ValidatedInternetAccess" value="true" />
<boolean name="NoInternetAccessExpected" value="false" />
<int name="UserApproved" value="0" />
<boolean name="MeteredHint" value="false" />
<int name="MeteredOverride" value="0" />
<boolean name="UseExternalScores" value="false" />
```

Figure 38: WifiConfigStore.xml: network Fios-pJK9q

```
<WifiConfiguration>
<string name="ConfigKey">&quot;TellMyWiFiLoveHer&quot;WPA_PSK</string>
<string name="SSID">&quot;TellMyWiFiLoveHer&quot;</string>
<null name="BSSID" />
<string name="PreSharedKey">&quot;cop576him662ill&quot;</string>
<int name="Priority" value="0" />
<null name="WEPKeys" />
<int name="WEPTxKeyIndex" value="0" />
<boolean name="HiddenSSID" value="false" />
<boolean name="RequirePMF" value="false" />
<byte-array name="AllowedKeyMgmt" num="1">02</byte-array>
<byte-array name="AllowedProtocols" num="1">03</byte-array>
<byte-array name="AllowedAuthAlgos" num="1">01</byte-array>
<byte-array name="AllowedGroupCiphers" num="1">0f</byte-array>
<byte-array name="AllowedPairwiseCiphers" num="1">06</byte-array>
<boolean name="Shared" value="true" />
<int name="Status" value="2" />
<null name="FQDN" />
<null name="ProviderFriendlyName" />
<null name="LinkedNetworksList" />
<null name="DefaultGwMacAddress" />
<boolean name="ValidatedInternetAccess" value="true" />
<boolean name="NoInternetAccessExpected" value="false" />
<int name="UserApproved" value="0" />
```

Figure 39: WifiConfigStore.xml: network TellMyWiFiLoveHer

Bluetooth. misc/bluedroid/bt_config.conf was created 2021-02-18 03:10:55. The adapter address is b0:f7:c4:41:cc:93 and its name is "Echo Show 8 (2nd Gen)-3SD". Two devices are paired:

MAC	Name in bt_config.conf	Other detail
f4:5c:89:c7:54:f0	Sarfraz's MacBook Pro	LinkKeyType 4; Bt_Metrics_Device_Connected = true; profiles a2dpsrc, hfpag
f4:4e:fd:c6:29:fc	onn. Rugged Speaker Mini	Timestamp 1696864570 = 2023-10-09 15:16:10 UTC; profiles a2dpsrc, a2dpsink, hfpclient

```
┌──(root㉿kali)-[/mnt/echoshow8-2]
└─# cd user_de/0

┌──(root㉿kali)-[/mnt/echoshow8-2/user_de/0]
└─# cd com.android.bluetooth

┌──(root㉿kali)-[/mnt/echoshow8-2/user_de/0/com.android.bluetooth]
└─# ls
cache  code_cache  lib  shared_prefs

┌──(root㉿kali)-[/mnt/echoshow8-2/user_de/0/com.android.bluetooth]
└─# cd shared_prefs

┌──(root㉿kali)-[/mnt/…/user_de/0/com.android.bluetooth/shared_prefs]
└─# ls
absolute_volume_blacklist.xml    Bt_Metrics_Remote_Device_Uuids.xml  message_access_permission.xml    sim_access_permission.xml
Bt_Metrics_Device_Connected.xml  Bt_Metrics_Shared_Pref.xml          phonebook_access_permission.xml  WirelessMetrics.xml

┌──(root㉿kali)-[/mnt/…/user_de/0/com.android.bluetooth/shared_prefs]
└─# cat Bt_Metrics_Device_Connected.xml
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<map>
    <boolean name="F4:5C:89:C7:54:F0" value="true" />
</map>

┌──(root㉿kali)-[/mnt/…/user_de/0/com.android.bluetooth/shared_prefs]
└─#
```

Figure 40: Bt_Metrics_Device_Connected.xml: MacBook MAC marked connected

```
┌──(root㉿kali)-[/mnt/…/user_de/0/com.android.bluetooth/shared_prefs]
└─# ls
absolute_volume_blacklist.xml   Bt_Metrics_Remote_Device_Uuids.xml  message_access_permission.xml    sim_access_permission.xml
Bt_Metrics_Device_Connected.xml  Bt_Metrics_Shared_Pref.xml         phonebook_access_permission.xml  WirelessMetrics.xml

┌──(root㉿kali)-[/mnt/…/user_de/0/com.android.bluetooth/shared_prefs]
└─# cat Bt_Metrics_Remote_Device_Uuids.xml
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<map>
    <string name="F4:5C:89:C7:54:F0">paired_dev_supp_uuids_a2dpsrc_hfpag_other</string>
    <string name="F4:4E:FD:C6:29:FC">paired_dev_supp_uuids_a2dpsrc_a2dpsink_hfpclient_other</string>
</map>

┌──(root㉿kali)-[/mnt/…/user_de/0/com.android.bluetooth/shared_prefs]
└─#
```

Figure 41: Bt_Metrics_Remote_Device_Uuids.xml: profiles of both paired devices

```
┌──(root㉿kali)-[/mnt/echoshow8-2]
└─# cd misc/bluedroid

┌──(root㉿kali)-[/mnt/echoshow8-2/misc/bluedroid]
└─# ls
bt_config.bak  bt_config.conf

┌──(root㉿kali)-[/mnt/echoshow8-2/misc/bluedroid]
└─# cat bt_config.conf
[Info]
FileSource = Empty
TimeCreated = 2021-02-18 03:10:55

[Adapter]
Address = b0:f7:c4:41:cc:93
LE_LOCAL_KEY_IRK = f2919fe4553d0315e5ef85fe95ddb68e
LE_LOCAL_KEY_IR = 2d98221ae17de4ace73d9b177b98570e
LE_LOCAL_KEY_DHK = 2752c4a667a0a9cfbe1b3b3c0feca007
LE_LOCAL_KEY_ER = edc6f1e8a7e16824644e8041ee543432
ScanMode = 1
DiscoveryTimeout = 1
Name = Echo Show 8 (2nd Gen)-3SD

[f4:5c:89:c7:54:f0]
Manufacturer = 0
LmpVer = 0
LmpSubVer = 0
Name = Sarfraz's MacBook Pro
```

Figure 42: bt_config.conf: adapter MAC and name Echo Show 8 (2nd Gen)-3SD

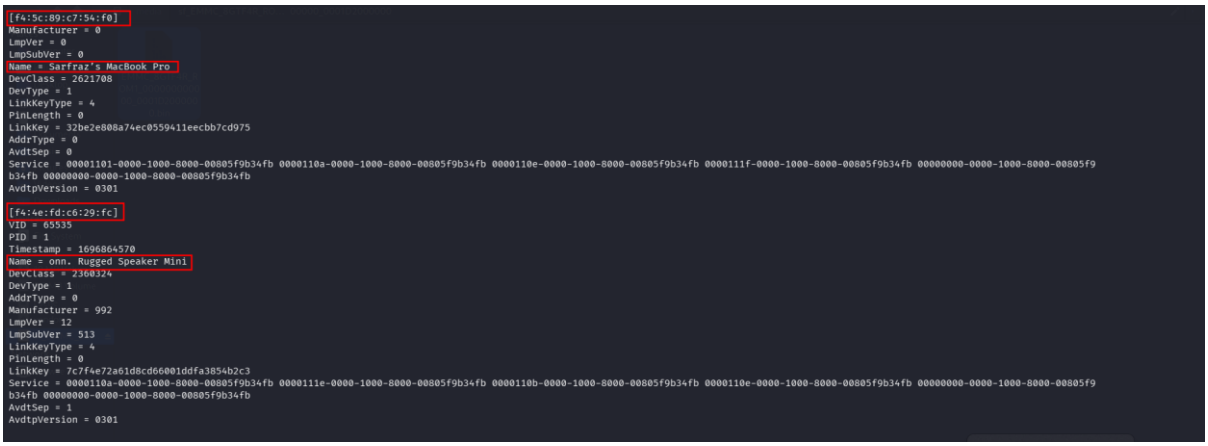


Figure 43: bt_config.conf: paired MacBook Pro and onn. speaker

Corroboration. In the bulk_extractor ether histogram, which comes from a raw scan of the image, the MacBook address is the most frequent MAC (305 uppercase + 176 lowercase hits). The speaker (144 + 39) and the Echo adapter (128 + 42) are also near the top (Figure 44). Snapshot 623 shows the same MacBook on the Bluetooth screen (Figure 36). Three independent sources agree. A device name such as "Sarfraz's MacBook Pro" is a label set on the laptop and does not prove who owned it.

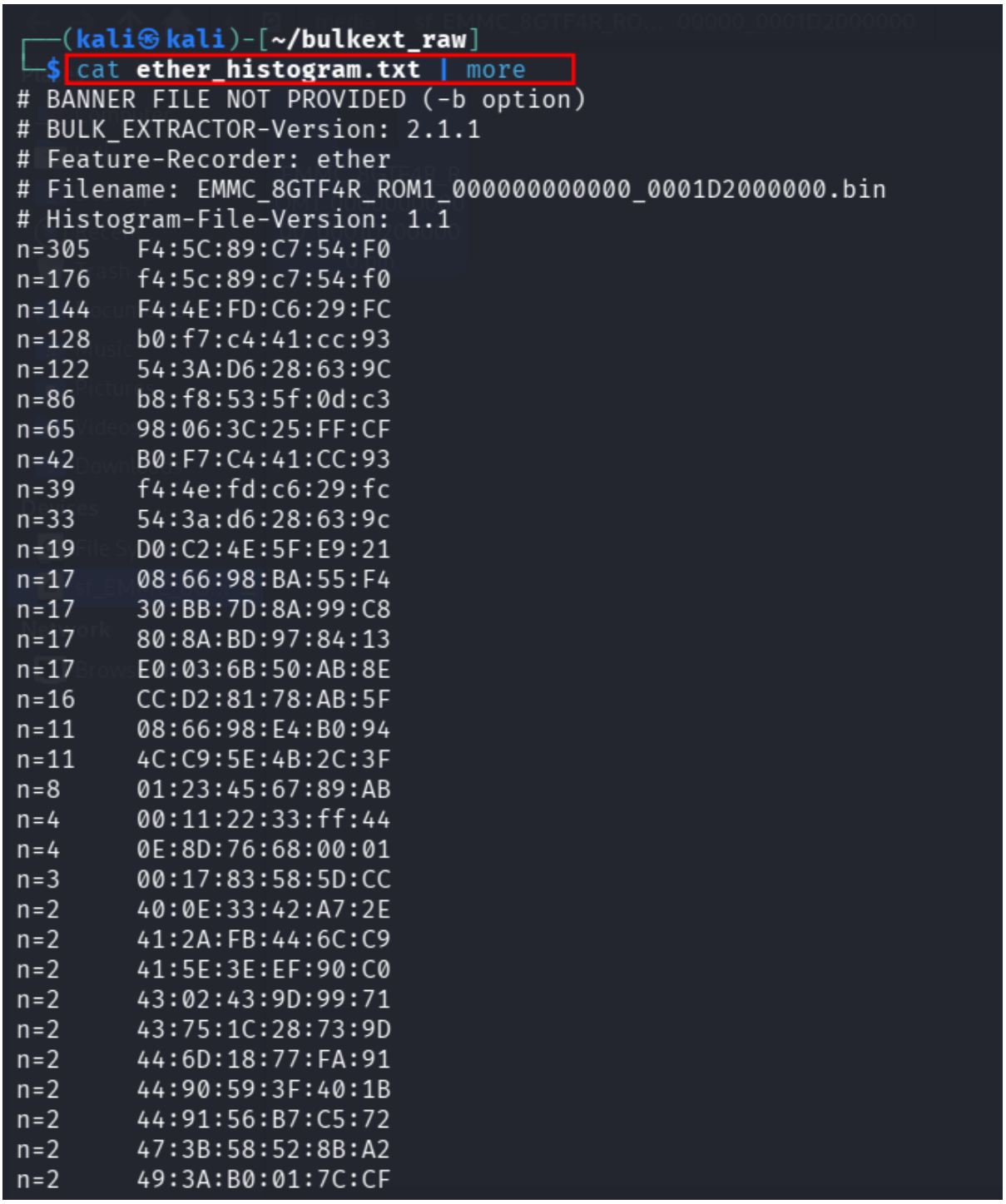


Figure 44: bulk_extractor ether (MAC address) histogram

Address data. data/com.amazon.kindle.otter.oobe/cache/volley/1111955081088707955 is a cached response from api.amazonalexa.com/v2/devices/~current/address-suggestions, dated Mon, 09 Oct 2023 20:40:08 GMT (X-Android-Received-Millis 1696884008711). It returns 3362 N Chatham Rd Apt C, Ellicott City, MD 21042-2752, US, with addressSource RETAIL_DEFAULT_SHIPPING and coordinates 39.279769, −76.833432 at accuracy 0.0 (Figure 45). This is the account's default shipping address sent by Amazon to the setup app, not a location fix from the device. A Maryland address fits the −0400 offset seen elsewhere.

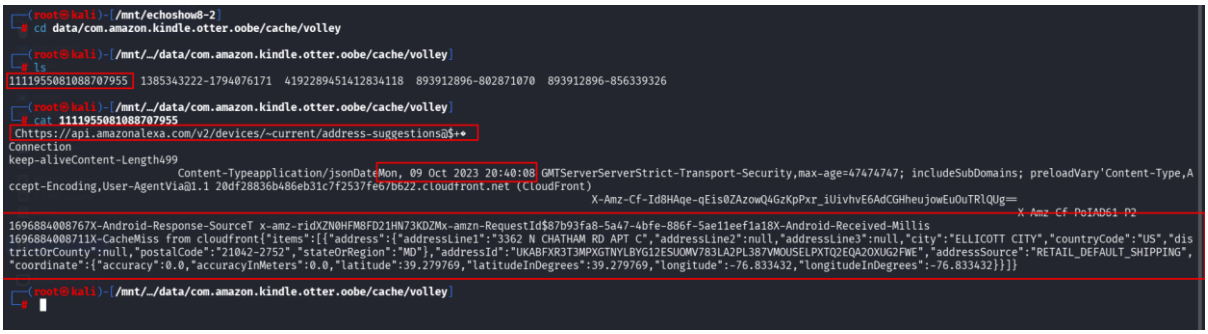


Figure 45: Cached Amazon address-suggestions response in the setup (OOBE) app

4.2 Web Activity

Source: data/com.amazon.cloud9/app_amazon_webview/amazon_webview/History, the Chromium profile used by the Silk web view (Figure 46). The visits table has 126 rows. The first visit is 13341364206672482 = 2023-10-09 22:30:06 UTC and the last is 13342495433605931 = 2023-10-23 00:43:53 UTC (Figures 47 and 48).

URL / title (urls table)	Raw last_visit_time	UTC
YouTube search "what's zero divided by zero" (ids 8, 9)	13341364595472780	2023-10-09 22:36:35
Generation Genius: Human Body Systems Video for Kids (id 40)	13341774395750811	2023-10-14 16:26:35
YouTube search "Kapil sharma" (id 50, visit_count 2)	13342240539770331	2023-10-20 01:55:39
Amazon: Alexa and Alexa Device FAQs (id 52)	13342400688326954	2023-10-21 22:24:48
login.live.com, username sarfy53@hotmail.com, OAuth for alexa.amazon.com 3PAccounts with Calendars and Contacts scopes (id 56)	13342408145145602	2023-10-22 00:29:05
"Account Linked" (ids 60 to 62)	13342408179474433	2023-10-22 00:29:39
YouTube search "kids videos" (id 63)	13342410629408781	2023-10-22 01:10:29
Google search "close google" (id 70)	13342492212049982	2023-10-22 23:50:12
Google search "search chicken nuggets on google" (id 69)	13342492250476128	2023-10-22 23:50:50

The keyword_search_terms table separately holds "search chicken nuggets on google" (url_id 69) and "close google" (url_id 70), which confirms those two searches (Figure 49). Their wording reads like spoken Alexa requests passed to the browser. That is an inference only. bulk_extractor also produced url_searches.txt and url_microsoft-live.txt from the raw image, which is an independent route to the same kind of records.

The Microsoft sign-in followed 34 seconds later by "Account Linked" shows a Microsoft account (sarfy53@hotmail.com) being linked to Alexa for calendar use. The calendar data itself would be held in the cloud, which is outside the scope of this examination. Visit times record when the web view loaded a page. They do not show who was in front of the screen.

```
┌──(root㉿kali)-[/mnt/echoshow8-2]
└─# cd data/com.amazon.cloud9/app_amazon_webview/amazon_webview

┌──(root㉿kali)-[/mnt/…/data/com.amazon.cloud9/app_amazon_webview/amazon_webview]
└─# ls
'Account Web Data'                Favicons                          'Network Action Predictor'                'Segmentation Platform'
'Account Web Data-journal'        Favicons-journal                  'Network Action Predictor-journal'        'Service Worker'
'Affiliation Database'            'Feature Engagement Tracker'      'Network Persistent State'                'Session Storage'
'Affiliation Database-journal'    feedv2                            'Offline Pages'                           shared_proto_db
 AutofillStrikeDatabase           'File System'                     optimization_guide_hint_cache_store       share_history
 blob_storage                     GPUCache                          optimization_guide_model_metadata_store   Shortcuts
 Bookmarks                        heavy_ad_intervention_opt_out.db  'Platform Notifications'                  Shortcuts-journal
 Bookmarks.bak                    heavy_ad_intervention_opt_out.db-journal  Preferences                       'Sync Data'
 BudgetDatabase                   History                           README                                    'Top Sites'
 commerce_subscription_db         History-journal                   RemoteConfiguration-aws_auth              'Top Sites-journal'
 Cookies                          IndexedDB                         RemoteConfiguration-experiments           TransportSecurity
 Cookies-journal                  'Local Storage'                   RemoteConfiguration-public-devices        VideoDecodeStats
 databases                        LOCK                              RemoteConfiguration-ua-override           'Visited Links'
 DawnCache                        LOG                               RemoteConfiguration-web-scaling-override  'Web Data'
 DIPS                             'Login Data'                      'Reporting and NEL'                       'Web Data-journal'
 DIPS-journal                     'Login Data-journal'              'Reporting and NEL-journal'               WebStorage
'Download Service'                LOG.old                           'SCT Auditing Pending Reports'

┌──(root㉿kali)-[/mnt/…/data/com.amazon.cloud9/app_amazon_webview/amazon_webview]
└─#
```

Figure 46: Web view profile folder for com.amazon.cloud9

```
┌──(root㉿kali)-[/mnt/…/data/com.amazon.cloud9/app_amazon_webview/amazon_webview]
└─# sqlite3 History
SQLite version 3.46.1 2024-08-13 09:16:08
Enter ".help" for usage hints.
sqlite> .tables
cluster_keywords          downloads                 segments
cluster_visit_duplicates  downloads_slices          typed_url_sync_metadata
clusters                  downloads_url_chains      urls
clusters_and_visits       keyword_search_terms      visit_source
content_annotations       meta                      visits
context_annotations       segment_usage
sqlite> .headers on
sqlite> SELECT * FROM visits;
id|url|visit_time|from_visit|transition|segment_id|visit_duration|incremented_omnibox_typed_score|opener_visit|originator_cache_guid|originator_visit_id|orig
inator_from_visit|originator_opener_visit|is_known_to_sync|consider_for_ntp_most_visited
1|1|13341364206672482|0|939524096|0|5285653|0|0||0|0|0|0|1
2|2|13341364211940733|0|805306368|0|156135153|0|1||0|0|0|0|1
3|3|13341364368056337|0|939524102|0|567358|0|0||0|0|0|0|1
4|4|13341364368604672|0|805306368|0|7115095|0|3||0|0|0|0|1
5|5|13341364375706195|0|805306368|0|107609043|0|4||0|0|0|0|1
6|4|13341364483312377|0|956301318|0|4542502|0|0||0|0|0|0|1
7|6|13341364487841868|0|805306368|0|45871514|0|6||0|0|0|0|1
8|7|13341364533696769|0|805306368|0|61791422|0|7||0|0|0|0|1
9|8|13341364595472780|0|939524102|0|542237|0|0||0|0|0|0|1
10|9|13341364595997613|0|805306368|0|1826376|0|9||0|0|0|0|1
11|7|13341364597820441|0|822083584|0|979862|0|0||0|0|0|0|1
12|4|13341364598798511|0|956301318|0|1501500|0|0||0|0|0|0|1
13|2|13341364600297498|0|822083584|0|4377891|0|0||0|0|0|0|1
```

Figure 47: History database tables and the start of the visits table

```
104|76|13342492270672197|103|2147483641|0|0|0|0||0|0|0|0|1
105|77|13342492270672197|104|2147483641|0|2059713|0|0||0|0|0|0|1
106|78|13342492272710881|105|1073741824|0|0|0|0||0|0|0|0|1
107|79|13342492272710881|106|1610612736|0|1093356|0|0||0|0|0|0|1
108|73|13342492273788830|107|805306375|0|91612554|0|0||0|0|0|0|1
109|80|13342492365395690|0|268435464|0|0|0|0||0|0|0|0|1
110|73|13342492365395690|109|1610612728|0|8445183|0|0||0|0|0|0|1
111|80|13342492373837536|0|268435464|0|0|0|0||0|0|0|0|1
112|73|13342492373837536|111|1610612728|0|1201375|0|0||0|0|0|0|1
113|80|13342492375035788|0|268435464|0|0|0|0||0|0|0|0|1
114|73|13342492375035788|113|1610612728|0|611722636|0|0||0|0|0|0|1
115|81|13342494925741702|0|939524102|0|1061842|0|0||0|0|0|0|1
116|82|13342494926790253|0|805306368|0|11944061|0|115||0|0|0|0|1
117|83|13342494938719330|0|805306368|0|0|0|116||0|0|0|0|1
118|81|13342495108804070|0|939524102|0|699441|0|0||0|0|0|0|1
119|82|13342495109492204|0|805306368|0|5149317|0|118||0|0|0|0|1
120|83|13342495114626577|0|805306368|0|2402350|0|119||0|0|0|0|1
121|83|13342495117015034|0|268435464|0|0|0|0||0|0|0|0|1
122|84|13342495117015034|121|1610612728|0|13635236|0|0||0|0|0|0|1
123|85|13342495130455156|0|805306368|0|199254903|0|122||0|0|0|0|1
124|86|13342495329668947|0|805306368|0|24001844|0|123||0|0|0|0|1
125|87|13342495353656843|0|805306368|0|79975366|0|124||0|0|0|0|1
126|88|13342495433605931|0|805306368|0|0|0|125||0|0|0|0|1
sqlite>
```

Figure 48: End of the visits table (126 rows)

```
┌──(root㉿kali)-[/mnt/…/data/com.amazon.cloud9/app_amazon_webview/amazon_webview]
└─# sqlite3 History
SQLite version 3.46.1 2024-08-13 09:16:08
Enter ".help" for usage hints.
sqlite> .headers on
sqlite> .tables
cluster_keywords          downloads                 segments
cluster_visit_duplicates  downloads_slices          typed_url_sync_metadata
clusters                  downloads_url_chains      urls
clusters_and_visits       keyword_search_terms      visit_source
content_annotations       meta                      visits
context_annotations       segment_usage
sqlite> SELECT * FROM keyword_search_terms;
keyword_id|url_id|term|normalized_term
3|69|search chicken nuggets on google|search chicken nuggets on google
3|70|close google|close google
sqlite>
```

Figure 49: keyword_search_terms: Google searches

```
┌──(root㉿kali)-[/mnt/…/data/com.amazon.cloud9/app_amazon_webview/amazon_webview]
└─# sqlite3 History
SQLite version 3.46.1 2024-08-13 09:16:08
Enter ".help" for usage hints.
sqlite> .header on
sqlite> SELECT * FROM urls;
id|url|title|visit_count|typed_count|last_visit_time|hidden
1|https://m.youtube.com/|Home - YouTube|5|0|13342240527071578|0
2|https://m.youtube.com/shorts/J89×5vmPB2A|Cooking Dinosaur!?! 🦕 - YouTube|2|0|13341364600297498|0
3|https://m.youtube.com/results?search_query=what%27s%20zero%20divided%20by%20one|YouTube|1|0|13341364368056337|0
4|https://m.youtube.com/results?search_query=what%27s+zero+divided+by+one|what's zero divided by one - YouTube|3|0|13341364598798511|0
5|https://m.youtube.com/watch?v=NKmGVE85GUU&pp=ygUad2hhdCdzIHplcm8gZGl2aWRlZCBieSBvbmU%3D|Why can't you divide by zero? - TED-Ed - YouTube|1|0|13341364375706195|0
6|https://m.youtube.com/shorts/dutekN9uuRw|Ask Siri what 0 divided by 0 is ... - YouTube|1|0|13341364487841868|0
7|https://m.youtube.com/shorts/gClVikU-iDs|YouTube|2|0|13341364597820441|0
8|https://m.youtube.com/results?search_query=what%27s%20zero%20divided%20by%20zero|YouTube|1|0|13341364595472780|0
9|https://m.youtube.com/results?search_query=what%27s+zero+divided+by+zero|what's zero divided by zero - YouTube|1|0|13341364595997613|0
10|https://m.youtube.com/shorts/8wxmPWkp8hU|#pov beetle juice tries to convince Lydia to say his name… #acting #beetlejuice - YouTube|1|0|13341364604645386|0
11|https://m.youtube.com/shorts/uUB2wpH2JgA|This girl starts to sing in the restaurant, EVERYONE WAS SHOCKED 😱 - YouTube|1|0|13341364644216062|0
12|https://m.youtube.com/shorts/umFENSW-zCM|I'm Inside Your House 😉 #shorts #funny #gaming - YouTube|1|0|13341364654727158|0
```

Figure 50: urls table: YouTube searches and videos

```
43|https://m.youtube.com/watch?v=bi5PhlIQpwU&pp=ygUQSG90IGluZGlhbiBtdXNpYw%3D%3D|Nadiyon Paar - Full Song |Roohi | Janhvi Kapoor| Sachin-Jigar | Shamur | Rashmeet K - YouTube|1|0|13342062940928082|0
44|https://m.youtube.com/watch?v=JFcgOboQZ08&pp=QAFIAQ%3D%3D|DILBAR Lyrical | Satyameva Jayate |John Abraham, Nora Fatehi,Tanishk B, Neha Kakkar,Dhvani, Ikka - YouTube|1|0|13342063133824057|0
45|https://m.youtube.com/watch?v=_uUdJalMaF8&pp=QAFIAg%3D%3D|Full Song: O SAKI SAKI | Batla House | Nora Fatehi, Tanishk B,Neha K,Tulsi K, B Praak,Vishal-Shekhar - YouTube|1|0|13342063343908518|0
46|https://m.youtube.com/watch?v=zqHUMF9syFA&pp=QAFIAw%3D%3D|Manike (Full Video): Thank God | Nora,Sidharth| Tanishk,Yohani,Jubin,Surya R |Rashmi Virag|Bhushan K - YouTube|1|0|13342063566649929|0
47|https://m.youtube.com/watch?v=RgzLnmTaCAU&pp=QAFIBA%3D%3D|Kusu Kusu Song Ft Nora Fatehi | Satyameva Jayate 2 | John A, Divya K | Tanishk B, Zahrah Khan, Dev N - YouTube|1|0|13342063817255990|0
48|https://m.youtube.com/watch?v=IuvVVWOsMBo&pp=QAFIBQ%3D%3D|Full Video: Ek Toh Kum Zindagani | Marjaavaan | Nora Fatehi | Tanishk B, Neha K, Yash N - YouTube|1|0|13342064052400275|0
49|https://m.youtube.com/watch?v=IE8OD5FbU-c&pp=QAFIBg%3D%3D|Full Song: Garmi | Street Dancer 3D | Varun D, Nora F, Badshah, Neha K | Remo D - YouTube|1|0|13342064272046532|0
50|https://m.youtube.com/results?sp=mAEA&search_query=Kapil+sharma|Kapil sharma - YouTube|2|0|13342240539770331|0
51|https://m.youtube.com/watch?v=p-ORaVBc5Zc&pp=ygUMS2FwaWwgc2hhcm1h|क्लासेस में Kapil द्वारा गया Yoga Instructor? | The Kapil Sharma Show Season 2 | Full Episode - YouTube|2|0|13342240542788213|0
52|https://www.amazon.com/gp/help/customer/display.html?nodeId=201602230|Alexa and Alexa Device FAQs - Amazon Customer Service|1|0|13342400688326954|0
53|https://login.microsoftonline.com/common/oauth2/v2.0/authorize?client_id=65ad87c3-6b07-40fa-93e6-e2a734bf2e9f&response_type=code&state=eyJ2YWxpZGF0aW9uQ29kZSI6ImhUb2dYK2Erc3ZPVDl2cThUbk5kam0xaDVYdWNzRmdRTHBxME00VzVGV0U9IiwiZGlyZWN0ZWRJZCI6IjhjYzVhYTVmLTEzNWEtOTNlZC1kYjllLTAwMmIxOGYxZTY0ZCIsInBhcnRuZXJJZCI6IndlclNJSWZkSjdhWWd2V1BxZG5mcDgzY0MiLCJhcHBsaWNhdGlvbkRvbWFpbiI6IkFMRVhBX0hITyIsImFwcGxpY2F0aW9uRG9tYWlucyI6WyJBTEVYQV9ISE8iXSwiZXhwaXJhdGlvblRpbWVJbk1pbGxpcyI6MTY5NzkzNzQyOTgzNSwidXNlclN0YXRlIjpbeyJrIjoiYWNjb3VudFVSUQiLCJ2IjoiOGNjNWFhNWYtMTM1YS05M2VkLWRiOWUtMDAyYjE4ZjFlNjRkIn0seyJrIjoicmVkaXJlY3RUb0FwcCIsInYiOiJmYWxzZSJ9LHsiayI6ImRldmljZUluaXRpYXRlZCIsInY0iJ0cnVlIn0seyJrIjoibGVnYWN5Q2FsZW5kYXJBY2NvdW50SWQiLCJ2IjpudWxsfSx7ImsiOiJsb2NhbGUiLCJ2IjoiZW4tdXMifSx7ImsiOiJpbml0aWF0aW5nUmVxdWVzdEIkIiwidiI6IjgzM2JhZmQxLTNlODUtNDI2Yy05ZTZhLTBmODRjZDFjMGYwMy9lLHsiayI6ImZsb3dUeXBlIiwidiI6IkxJTktJTkcifV0sImRpcmVjdGVkSWRUeXBlIjoiVVVJRCIsImdyYW50ZWRTY29wZXNJbmNsdWRlZCI6ZmFsc2V9&scope=https%3A%2F%2Fgraph.microsoft.com%2FPeople.Read+https%3A%2F%2Fgraph.microsoft.com%2FContacts.Read+offline_access+https%3A%2F%2Fgraph.microsoft.com%2FCalendars.Read.Shared+https%3A%2F%2Fgraph.microsoft.com%2FUser.Read+https%3A%2F%2Fgraph.microsoft.com%2FCalendars.ReadWrite+https%3A%2F%2Fgraph.microsoft.com%2FCalendars.ReadWrite.Shared&redirect_uri=https%3A%2F%2Falexa.amazon.com%2Fapi%2F3PAccounts%2Foauth%2FestablishAccess&prompt=select_account|Sign in to your account|1|0|13342407432028665|0
```

Figure 51: urls table: Kapil Sharma search, Alexa FAQ and Microsoft OAuth

```
N1%26uaid%3d0b1ceb5b95c84f158c415a49d7f278ab%26client_id%3d65ad87c3-6b07-40fa-93e6-e2a734bf2e9f%26opid%3d1D443372257DCC7C%26mkt%3dEN-US%26opidt%3d1697934565|
Let this app access your info? (1 of 1 apps)|1|0|13342408165962020|0
60|https://login.live.com/oauth20_authorize.srf?route=C107_BAY&uaid=0b1ceb5b95c84f158c415a49d7f278ab&client_id=65ad87c3-6b07-40fa-93e6-e2a734bf2e9f&opid=1D443372257DCC7C&mkt=EN-US&opidt=1697934565&res=success|Account Linked|1|0|13342408179474433|0
61|https://alexa.amazon.com/api/3PAccounts/oauth/establishAccess?code=M.C107_BAY.2.0a7bd2bf-d5b2-4262-85ab-99260831117a&state=eyJ2YWxpZGF0aW9uQ29kZSI6ImV3Y3FKRXE2Z3NuWkprODl2VVUzdFlhYWlWUUxsZDFwMGRnamxnRHlqdmM9IiwiZGlyZWN0ZWRJZCI6IjM0YzVhYTY0LTVlY2EtNzUzMi1lYjQxLWRkMDRhODA1OTliMiIsInBhcnRuZXJJZCI6IndlclNJSWZkSjdhWWd2V1BxZG5mcDgzY0MiLCJhcHBsaWNhdGlvbkRvbWFpbiI6IkFMRVhBX0hITyIsImFwcGxpY2F0aW9uRG9tYWlucyI6WyJBTEVYQV9ISE8iXSwiZXhwaXJhdGlvblRpbWVJbk1pbGxpcyI6MTY5NzkzODEyMzgxNywidXNlclN0YXRlIjpbeyJrIjoiYWNjb3VudFVSUQiLCJ2IjoiMzRjNWFhNjQtNWVjYS03NTMyLWViNDEtZGQwNGE4MDU5OWIyIn0seyJrIjoicmVkaXJlY3RUb0FwcCIsInYiOiJmYWxzZSJ9LHsiayI6ImRldmljZUluaXRpYXRlZCIsInY0iJ0cnVlIn0seyJrIjoibGVnYWN5Q2FsZW5kYXJBY2NvdW50SWQiLCJ2IjpudWxsfSx7ImsiOiJsb2NhbGUiLCJ2IjoiZW4tdXMifSx7ImsiOiJpbml0aWF0aW5nUmVxdWVzdEIkIiwidiI6IjY4NDk0Nzl0LTkwNDItNGNmYy1iM2E0LWY1Yjk0ODhjMTczNSJ9LHsiayI6IkxJTktJTkcifV0sImRpcmVjdGVkSWRUeXBlIjoiVVVJRCIsImdyYW50ZWRTY29wZXNJbmNsdWRlZCI6ZmFsc2V9|Account Linked|1|0|13342408179474433|0
62|https://account-linking-result.com/en-us/success.html|Account Linked|1|0|13342408179474433|0
63|https://m.youtube.com/results?search_query=kids%20videos|YouTube|1|0|13342410629408781|0
64|https://m.youtube.com/results?search_query=kids+videos|kids videos - YouTube|1|0|13342410630863493|0
65|http://bing.com/|Bing|1|0|13342492001629428|0
66|https://bing.com/|Bing|1|0|13342492001629428|0
67|https://www.bing.com/?toWww=1&redig=FDE7754DDDD74D53AAC684BAF5E19FB3|Bing|2|0|13342492255418343|0
68|https://www.google.com/|Google|2|0|13342492252941224|0
69|https://www.google.com/search?q=search%20chicken%20nuggets%20on%20google|search chicken nuggets on google - Google Search|2|0|13342492250476128|0
70|https://www.google.com/search?q=close%20google|close google - Google Search|1|0|13342492212049982|0
71|https://account.microsoft.com/|Continue|1|0|13342492268854145|0
72|https://login.live.com/login.srf?wa=wsignin1.0&rpsnv=16&checkda=1&ct=1698018668&rver=7.5.2112.0&wp=MBI_SSL&wreply=https:%
```

Figure 52: urls table: Account Linked, kids videos, Bing and Google searches

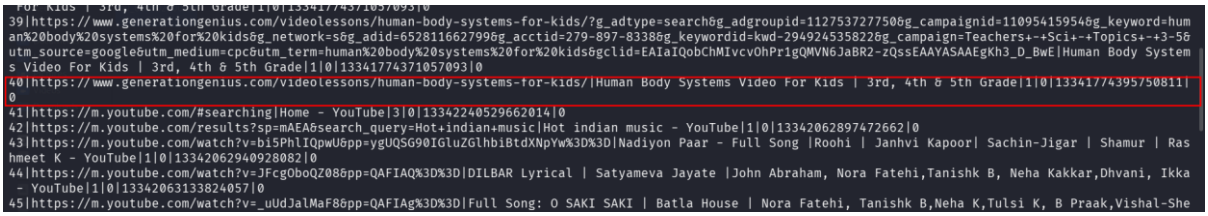


Figure 53: urls table: Generation Genius and YouTube music searches


55|https://login.microsoftonline.com/common/oauth2/v2.0/authorize?client_id=65ad87c3-6b07-40fa-93e6-e2a734bf2e9f&response_type=code&state=eyJ2YWxpZGF0aW9uQ29kZSI6ImV3Y3FKRXE2Z3NuWkprODl2VVUzdFlhYWlWUUxsZDFwMGRnamxnRHlqdmM9IiwiZGlyZWN0ZWRJZCI6IjM0YzVhYTY0LTVlY2EtNzUzMi1lYjQxLWRkMDRhODA1OTliMiIsInBhcnRuZXJJZCI6IndlclNJSWZkSjdhWWd2V1BxZG5mcDgzY0MiLCJhcHBsaWNhdGlvbkRvbWFpbiI6IkFMRVhBX0hITyIsImFwcGxpY2F0aW9uRG9tYWlucyI6WyJBTEVYQV9ISE8iXSwiZXhwaXJhdGlvblRpbWVJbk1pbGxpcyI6MTY5NzkzODEyMzgxNywidXNlclN0YXRlIjpbeyJrIjoiYWNjb3VudFVVSUQiLCJ2IjoiMzRjNWFhNjQtNWVjYS03NTMyLWViNDEtZGQwNGE4MDU5OWIyIn0seyJrIjoicmVkaXJlY3RUb0FwcCIsInYiOiJmYWxzZSJ9LHsiayI6ImRldmljZUluaXRpYXRlZCIsInYiOiJ0cnVlIn0seyJrIjoibGVnYWN5Q2FsZW5kYXJBY2NvdW50SWQiLCJ2IjpudWxsfSx7ImsiOiJsb2NhbGUiLCJ2IjoiZW4tdXMifSx7ImsiOiJpbml0aWF0aW5nUmVxdWVzdEVsYXBzZWRUaW1lIiwidiI6IjY4NDk0NzI0LTkwNDItNGNmYy1iM0ZlLWE1MjdkMWY

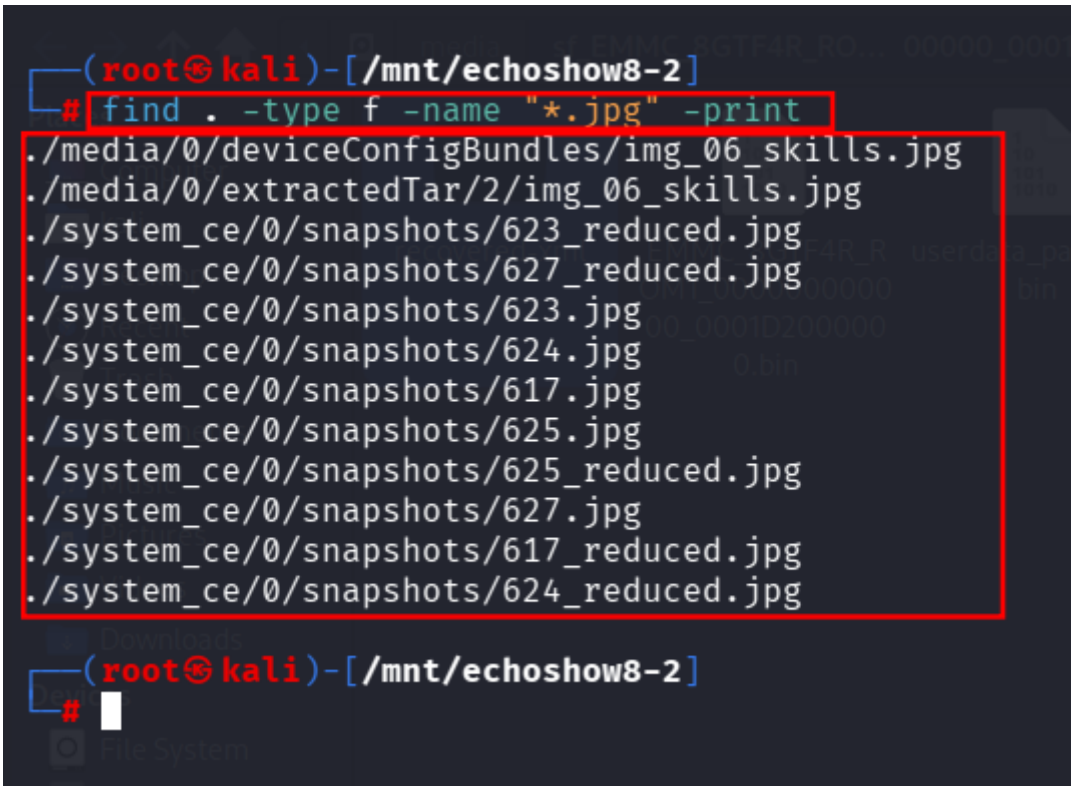


Figure 55: JPG files on the userdata partition



Figure 56: No JPG files on the system partition

Photo Booth uploads.

data/com.amazon.zordon/databases/amzn1.account.AFTJBUB4CIBBMH2S2AQDEY5QCN2A.mixtape.db belongs to the Amazon Photos service. Its upload_requests table has eight rows, all in state DONE, each with a file path, size and MD5 (Figures 57 and 58). The cloud_nodes table adds the creation date and camera fields: make Amazon, model AEOAT, exposure 1/120, no flash. So the device's own camera took these photos (Figure 59).

File (data/user/0/com.amazon.zordon/files/)	Size (bytes)	cloud_nodes created_date (UTC)
photo_booth_single_shot2023-10-02_17-52-26-169.jpeg	2,849,322	1696283551557 = 2023-10-02 21:52:31
photo_booth_single_shot2023-10-07_13-25-13-104.jpeg	3,010,907	1696699520501 = 2023-10-07 17:25:20
photo_booth_single_shot2023-10-09_15-34-15-361.jpeg	3,353,396	1696880061012 = 2023-10-09 19:34:21
photo_booth_video1696880081230.mp4	6,047,662	1696880095819 = 2023-10-09 19:34:55
photo_booth_single_shot2023-10-09_19-45-07-628.jpeg	2,935,516	1696895111692 = 2023-10-09 23:45:11
photo_booth_single_shot2023-10-17_20-29-28-381.jpeg	2,717,919	1697588974960 = 2023-10-18 00:29:34
photo_booth_single_shot2023-10-21_19-02-39-687.jpeg	2,961,976	1697929372480 = 2023-10-21 23:02:52
photo_booth_video1697929429134.mp4	13,958,448	1697929464021 = 2023-10-21 23:04:24

Every photo filename sits four hours and a few seconds before its UTC creation record. The filenames therefore use local EDT, and each upload record followed the capture within seconds. The image_date_time_original field repeats the local time as if it were UTC (for example 1696269146000 for the 2 October photo), which is a common trap when reading these values. The video filenames hold true epoch milliseconds: 1697929429134 = 2023-10-21 23:03:49 UTC, 35 seconds before its upload record.

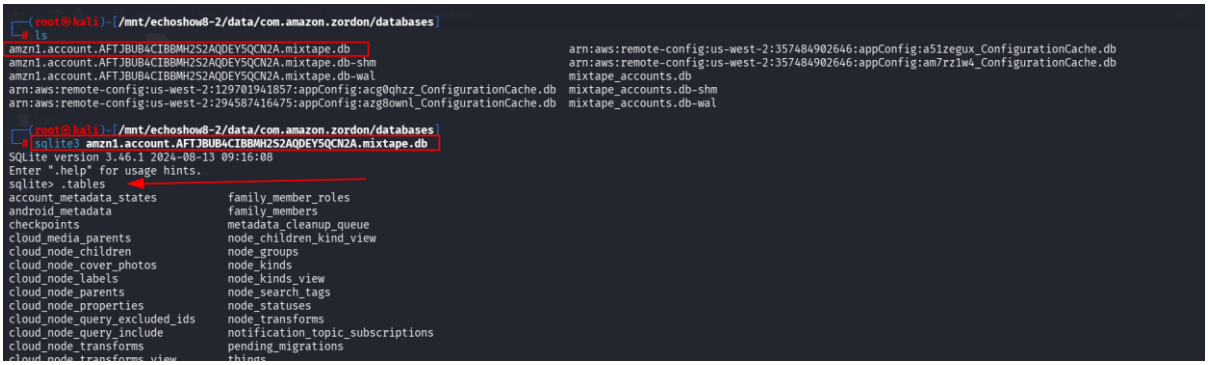


Figure 57: Zordon (Amazon Photos) databases and tables

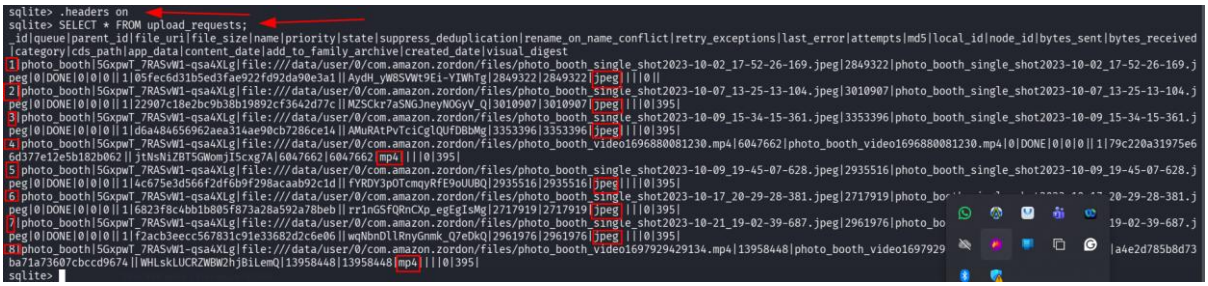


Figure 58: upload_requests: six Photo Booth photos and two videos, all DONE


sqlite> SELECT * FROM cloud_nodes;
_id|event_id|node_id|created_by|created_date|description|exclusively_trashed|is_root|is_shared|kind_id|modified_date|name|recursively_trashed|restricted|status_id|version|content_type|content_extension|content_md5|content_visual_digest|content_sort_date|content_date|content_size|content_version|document_version|document_title|image_aperture_value|image_capture_mode|image_color_space|image_date_time|image_date_time_digitized|image_date_time_original|image_exposure_mode|image_exposure_program|image_exposure_time|image_flash|image_focal_length|image_gps_time_stamp|image_height|image_iso_speed_ratings|image_location|image_make|image_metering_mode|image_model|image_orientation|image_resolution_unit|image_sensing_method|image_sharpness|image_software|image_white_balance|image_width|image_x_resolution|image_y_resolution|is_content|is_document|is_image|is_video|video_audio_bitrate|video_audio_channels|video_audio_channel_layout|video_audio_codec|video_audio_sample_rate|video_bitrate|video_creation_date|video_duration|video_encoder|video_height|video_location|video_make|video_model|video_rotate|video_title|video_overall_bitrate|video_codec|video_frame_rate|video_width|favorite|hidden|metadata_version|has_share_parent|family_member_id
1|1|AydH_yW8SVWt9Ei-YIWhTg||1696283551557|||||1|1696283552715|photo_booth_single_shot2023-10-02_17-52-26-169.jpeg||0|1|2|image/jpeg|jpg|05fec6d31b5ed3fae922fd92da90e3a1|64c52dcb95afb12fd1ad474ed9195f27|1696283546015|1696269146015|2849322|1|||||sRGB|1696269146000|1696269146000|1696269146000|Auto|Not Defined|1/120|No Flash|3.8||2160|||Amazon|Center-weighted average|AEOAT|1|Pixels/Inch|||custom software|Auto|3456|||1|0|1|0|||||||||||||||||||0|0|5|0|–1
2|2|MZSCkr7aSNGJneyNOGyV_Q||1696699520501|||||1|1696699521889|photo_booth_single_shot2023-10-07_13-25-13-104.jpeg||0|1|2|image/jpeg|jpg|22907c18e2bc9b38b19892cf3642d77c|92e39c4a10b616f83c81eabefdf51e30|1696699513094|1696685113094|3010907|1|||||sRGB|1696685113000|1696685113000|1696685113000|Auto|Not Defined|1/120|No Flash|3.8||2160|||Amazon|Center-weighted average|AEOAT|1|Pixels/Inch|||custom software|Auto|3456|||1|0|1|0|||||||||||||||||||0|0|5|0|–1
3|3|AMuRAtPvTciCglQUfDBbMg||1696880061012|||||1|1696880062078|photo_booth_single_shot2023-10-09_15-34-15-361.jpeg||0|1|2|image/jpeg|jpg|d6a484656962aea314ae90cb7286ce14|9b952465994204e06f43c178508872e9|1696880055034|1696865655034|3353396|1|||||sRGB|1696865655000|1696865655000|1696865655000|Auto|Not Defined|1/120|No Flash|3.8||2160|||Amazon|Center-weighted average|AEOAT|1|Pixels/Inch|||custom software|Auto|3456|||1|0|1|0|||||||||||||||||||0|0|5|0|–1
4|4|jtNsNiZBT5GWomjI5cxg7A||1696880095819|||||1|1696880097618|photo_booth_video1696880081230.mp4||0|1|3|video/mp4|mp4|79c220a31975e66d377e12e5b182b062||1696880095819||6047662|1|||||||||||||||||||



Figure 60: Conversion of two cloud_nodes created_date values

Carving. I ran scalpel 1.60 on userdata_partition.bin with JPG and an XML rule (header <?xml, 20,000 bytes) enabled. It built carve lists of 12 + 6,296 JPG candidates and 12,205 XML candidates, and its audit file lists each carved file with its offset (Figures 61 to 65). I then ran foremost 1.5.7 for JPG over the whole image (Figures 68 to 71). It extracted 6,390 JPG files in 12 m 21 s, many of them repeated 163 KB and 174 KB copies of the same interface images.



Figure 61: Editing the scalpel configuration

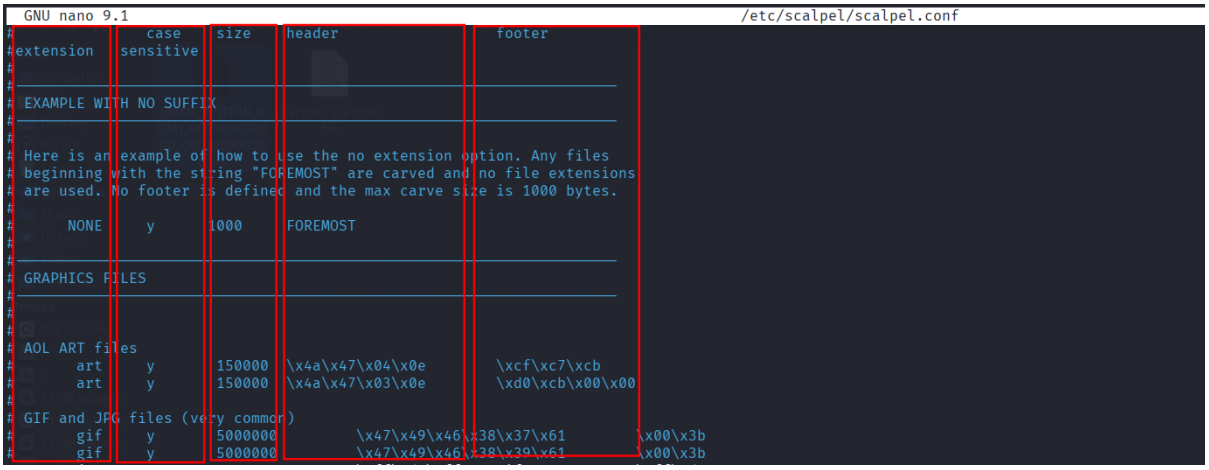


Figure 62: scalpel.conf: rule columns (extension, case, size, header, footer)

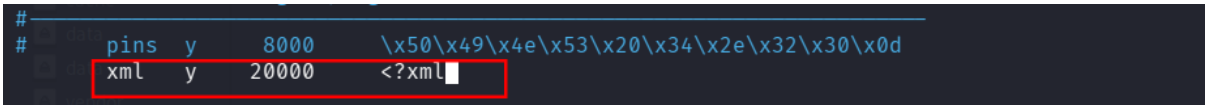


Figure 63: scalpel.conf: XML rule added



Figure 64: scalpel run on userdata_partition.bin

```
┌──(kali㉿kali)-[/media/sf_EMMC_8GTF4R_ROM1_000000000000_0001D2000000]
└─$ cd recovered-xml

┌──(kali㉿kali)-[/media/sf_EMMC_8GTF4R_ROM1_000000000000_0001D2000000/recovered-xml]
└─$ ls
audit.txt  jpg-1-0  jpg-1-2  jpg-1-4  jpg-1-6  xml-2-1   xml-2-11  xml-2-2  xml-2-4  xml-2-6  xml-2-8
jpg-0-0    jpg-1-1  jpg-1-3  jpg-1-5  xml-2-0  xml-2-10  xml-2-12  xml-2-3  xml-2-5  xml-2-7  xml-2-9

┌──(kali㉿kali)-[/media/sf_EMMC_8GTF4R_ROM1_000000000000_0001D2000000/recovered-xml]
└─$ cat audit.txt

Scalpel version 1.60 audit file
Started at Sun Oct 18 05:32:54 2026
Command line:
scalpel -o recovered-xml userdata_partition.bin

Output directory: /media/sf_EMMC_8GTF4R_ROM1_000000000000_0001D2000000/recovered-xml
Configuration file: /etc/scalpel/scalpel.conf

Opening target "/media/sf_EMMC_8GTF4R_ROM1_000000000000_0001D2000000/userdata_partition.bin"

The following files were carved:
File             Start          Chop          Length         Extracted From
00000013.jpg     10428416       NO          3852744          userdata_partition.bin
00000012.jpg      9908224       NO          4372936          userdata_partition.bin
00000014.jpg     14065664       NO          5230914          userdata_partition.bin
00006308.xml     12582912       YES           20000          userdata_partition.bin
00006309.xml    449131116       YES           20000          userdata_partition.bin
00006315.xml    529514256       YES           20000          userdata_partition.bin
00006314.xml    529487998       YES           20000          userdata_partition.bin
00006313.xml    529484969       YES           20000          userdata_partition.bin
00006312.xml    529484630       YES           20000          userdata_partition.bin
00006311.xml    529461363       YES           20000          userdata_partition.bin
00006310.xml    529447968       YES           20000          userdata_partition.bin
```

Figure 65: scalpel output folders and audit file

```
┌──(root㉿kali)-[/mnt/echoshow8]
└─# df -h
Filesystem                                      Size  Used Avail Use% Mounted on
udev                                            2.2G     0  2.2G   0% /dev
tmpfs                                           457M  1.3M  456M   1% /run
/dev/sda1                                        79G   45G   31G  60% /
tmpfs                                           2.3G  4.0K  2.3G   1% /dev/shm
none                                            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs                                           2.3G  8.0K  2.3G   1% /tmp
none                                            1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
none                                            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs                                           457M  108K  457M   1% /run/user/1000
EMMC_8GTF4R_ROM1_000000000000_0001D2000000      953G  948G  5.7G 100% /media/sf_EMMC_8GTF4R_ROM1_000000000000_0001D2000000
/dev/loop0p23                                   3.4G  2.3G  1.1G  68% /mnt/echoshow8-2
/dev/loop0p21                                   3.0G  2.0G  978M  68% /mnt/echoshow8

┌──(root㉿kali)-[/mnt/echoshow8]
└─#
```

Figure 66: Free space check before carving

```
# ---------------------------------------------------------------
# GRAPHICS FILES
# ---------------------------------------------------------------
#
#
# AOL ART files
#       art     y       150000  \x4a\x47\x04\x0e        \xcf\xc7\xcb
#       art     y       150000  \x4a\x47\x03\x0e        \xd0\xcb\x00\x00
#
# GIF and JPG files (very common)
#       gif     y       5000000         \x47\x49\x46\x38\x37\x61        \x00\x3b
#       gif     y       5000000         \x47\x49\x46\x38\x39\x61        \x00\x3b
#       jpg     y       5242880         \xff\xd8\xff???Exif             \xff\xd9        REVERSE
        jpg     y       5242880         \xff\xd8\xff???JFIF             \xff\xd9        REVERSE
#
#
# PNG
#       png     y       20000000        \x50\x4e\x47?   \xff\xfc\xfd\xfe
#
#
# BMP   (used by MSWindows, use only if you have reason to think there are
#       BMP files worth digging for. This often kicks back a lot of false
#       positives
```

Figure 67: scalpel.conf: JPG rule enabled

```
┌──(kali㉿kali)-[/media/sf_EMMC_8GTF4R_ROM1_000000000000_0001D2000000]
└─$ foremost -t jpg -i /media/sf_EMMC_8GTF4R_ROM1_000000000000_0001D2000000/EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin -o ~/foremost-jpg
Processing: /media/sf_EMMC_8GTF4R_ROM1_000000000000_0001D2000000/EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin
|*****************************************************************|

┌──(kali㉿kali)-[/media/sf_EMMC_8GTF4R_ROM1_000000000000_0001D2000000]
└─$
```

Figure 68: foremost JPG carve of the full image

```
┌──(kali㉿kali)-[/media/sf_EMMC_8GTF4R_ROM1_000000000000_0001D2000000]
└─$ cd ~/foremost-jpg

┌──(kali㉿kali)-[~/foremost-jpg]
└─$ ls
audit.txt  jpg

┌──(kali㉿kali)-[~/foremost-jpg]
└─$ cd jpg

┌──(kali㉿kali)-[~/foremost-jpg/jpg]
└─$ ll |more
total 173716
-rw-rw-r-- 1 kali kali  167019 Oct 18 09:58 01271252.jpg
-rw-rw-r-- 1 kali kali  178820 Oct 18 09:58 01272327.jpg
-rw-rw-r-- 1 kali kali  167019 Oct 18 09:58 01287629.jpg
-rw-rw-r-- 1 kali kali  178820 Oct 18 09:58 01288368.jpg
-rw-rw-r-- 1 kali kali    1989 Oct 18 09:59 02800818.jpg
-rw-rw-r-- 1 kali kali    3273 Oct 18 09:59 02800828.jpg
-rw-rw-r-- 1 kali kali  167019 Oct 18 09:59 02813821.jpg
-rw-rw-r-- 1 kali kali  178820 Oct 18 09:59 02814560.jpg
-rw-rw-r-- 1 kali kali  167019 Oct 18 09:59 02864092.jpg
-rw-rw-r-- 1 kali kali  178820 Oct 18 09:59 02864907.jpg
-rw-rw-r-- 1 kali kali  167019 Oct 18 09:59 02911553.jpg
-rw-rw-r-- 1 kali kali  178820 Oct 18 09:59 02912886.jpg
```

Figure 69: foremost carved JPG files

```
┌──(kali㉿kali)-[~/foremost-jpg/jpg]
└─$ cd ..

┌──(kali㉿kali)-[~/foremost-jpg]
└─$ ls
audit.txt  jpg

┌──(kali㉿kali)-[~/foremost-jpg]
└─$ cat audit.txt
Foremost version 1.5.7 by Jesse Kornblum, Kris Kendall, and Nick Mikus
Audit File

Foremost started at Sun Oct 18 09:56:54 2026
Invocation: foremost -t jpg -i /media/sf_EMMC_8GTF4R_ROM1_000000000000_0001D2000000/EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin -o /home/kali/foremost-jpg
Output directory: /home/kali/foremost-jpg
Configuration file: /etc/foremost.conf
------------------------------------------------------------------
File: /media/sf_EMMC_8GTF4R_ROM1_000000000000_0001D2000000/EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin
Start: Sun Oct 18 09:56:54 2026
Length: 7 GB (7818182656 bytes)

Num      Name (bs=512)         Size      File Offset     Comment

0:      01271252.jpg         163 KB       650881084
1:      01272327.jpg         174 KB       651431844
2:      01287629.jpg         163 KB       659266088
3:      01288368.jpg         174 KB       659644812
4:      02800818.jpg           1 KB      1434018976
5:      02800828.jpg           3 KB      1434023960
6:      02813821.jpg         163 KB      1440676760
7:      02814560.jpg         174 KB      1441054896
8:      02864092.jpg         163 KB      1466415248
9:      02864907.jpg         174 KB      1466832620
10:     02911553.jpg         163 KB      1490715632
11:     02912886.jpg         174 KB      1491397896
12:     03259030.jpg          62 KB      1668623756
13:     03259156.jpg          60 KB      1668688136
```

Figure 70: foremost audit: start time, image length and file offsets

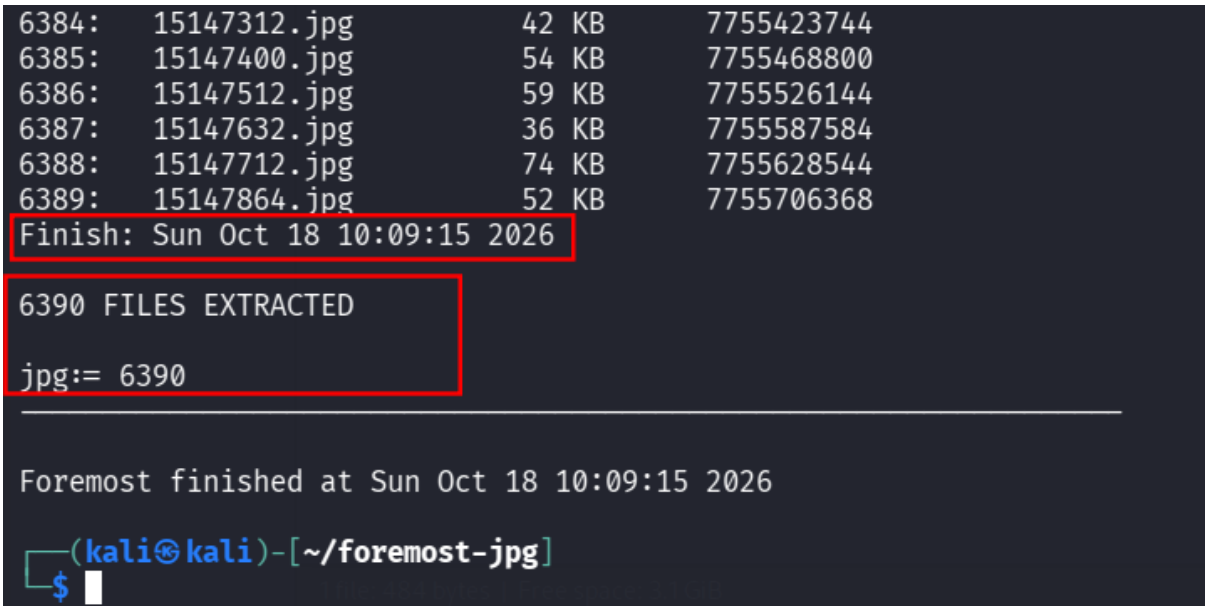


Figure 71: foremost audit: 6,390 JPG files extracted

Sample carved images include an outdoor nature photo and three dark film-style stills (Figures 72 to 75). The stills look like video artwork or thumbnails cached by an app, but nothing in the carved files confirms their source. ExifTool on 01271252.jpg shows a 480 × 480 JFIF image with no EXIF block. Its file dates (2026-10-18 09:58 −04:00) are the time foremost wrote the file, not the original capture time (Figures 76 and 77).

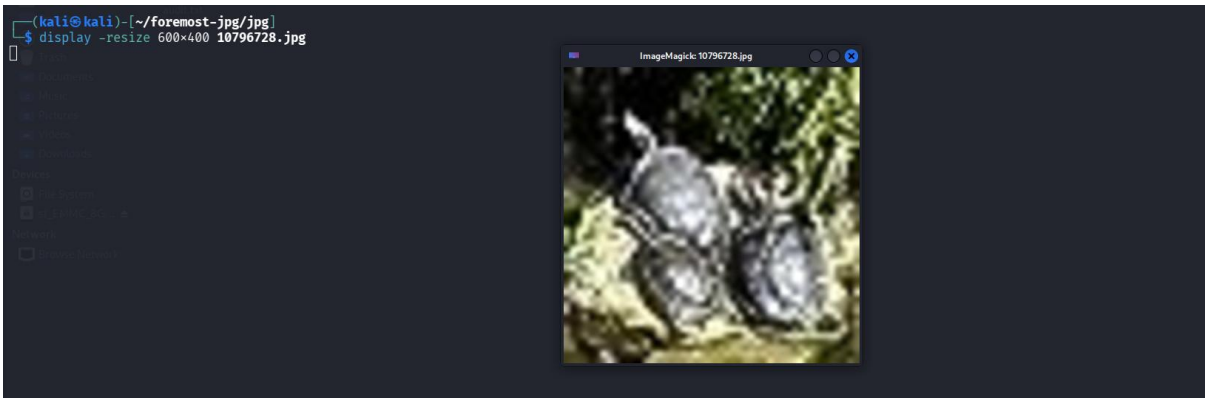


Figure 72: Carved image 10796728.jpg

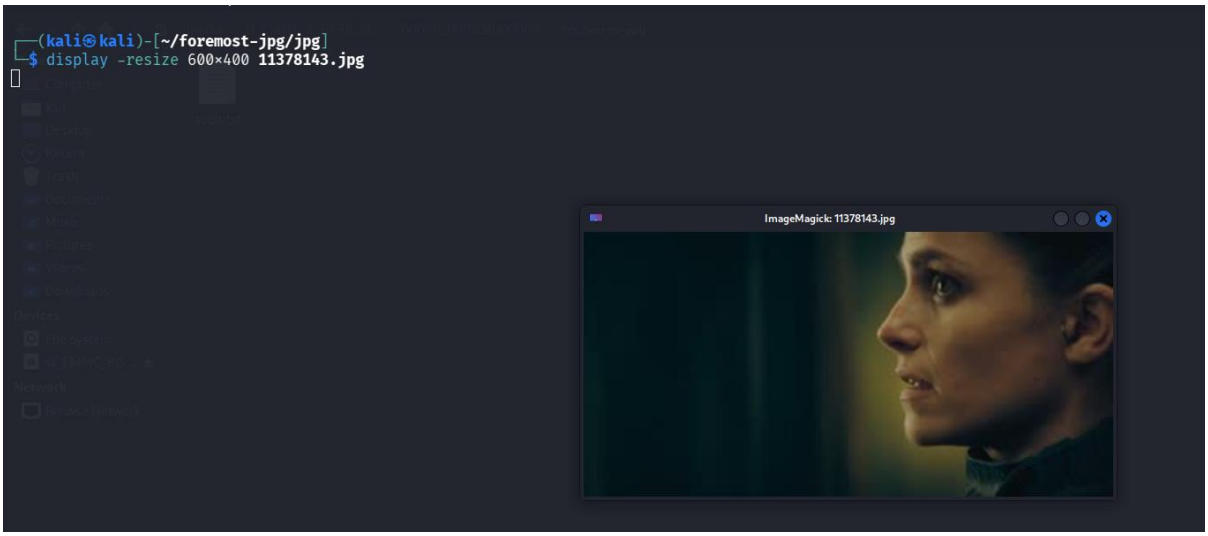


Figure 73: Carved image 11378143.jpg

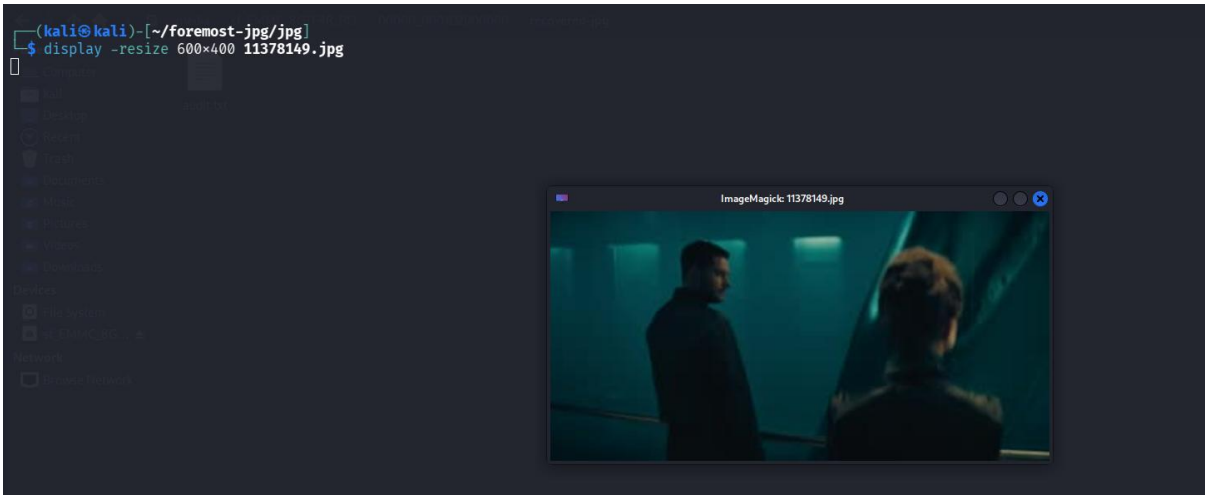


Figure 74: Carved image 11378149.jpg

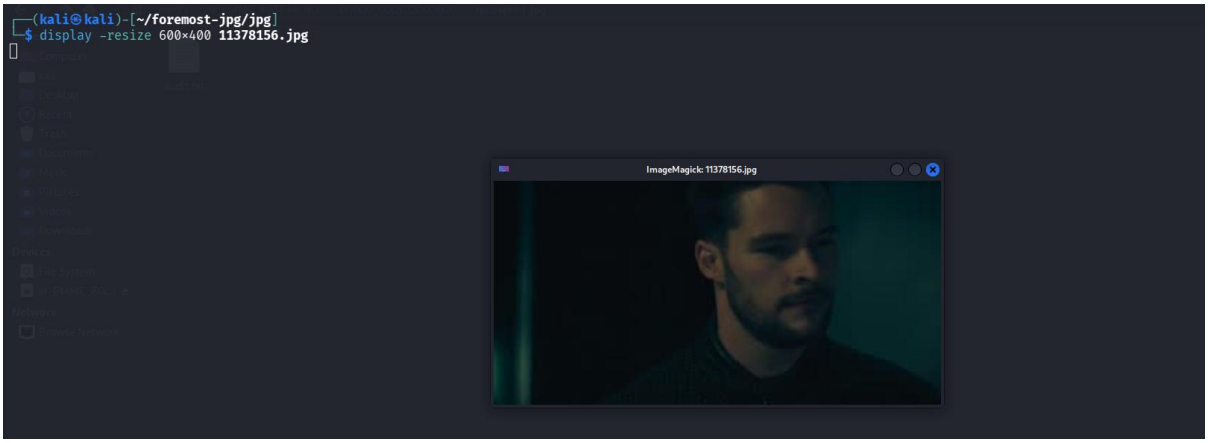


Figure 75: Carved image 11378156.jpg

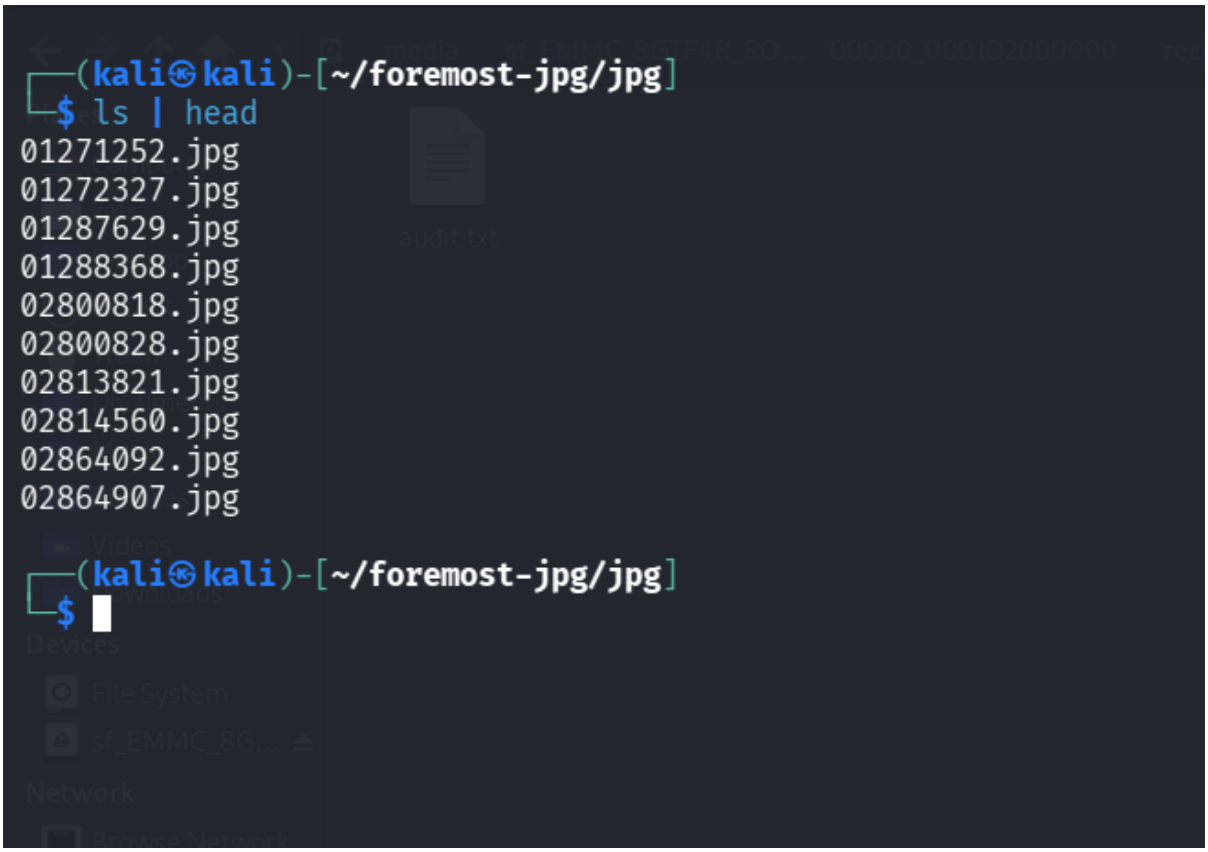


Figure 76: Carved JPG file names

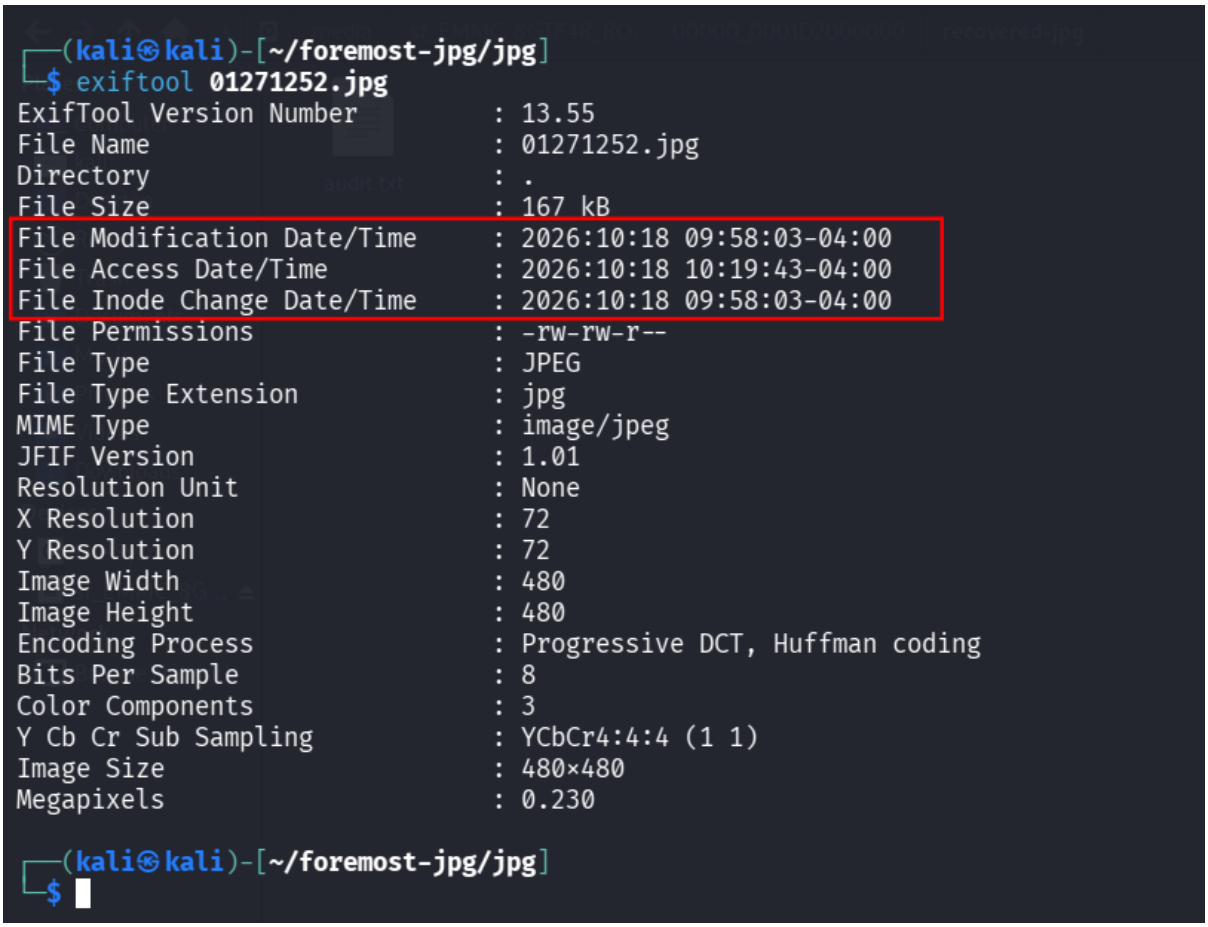


Figure 77: ExifTool output for a carved JPG: no EXIF, examiner-time file dates

4.5 Videos

Prime Video. I copied data/com.amazon.avod/files/databases/dbplaybackhistory (with its other databases) to /home/kali/case/avod and queried playbackHistory (Figures 78 and 79). All five rows use account amzn1.account.AFTJBUB4CIBBMH2S2AQDEY5QCN2A and profile amzn1.actor.person.oid.A2YI4Z0W6FQDFA.

Title	Type	lastAccessed (raw)	UTC	Position / runtime
Top Gun: Maverick	Movie, PG-13	1698016149065	2023-10-22 23:09:09	1:42:27 / 2:10:14
Jurassic World Dominion	Movie, PG-13	1697932011023	2023-10-21 23:46:51	0:01:46 / 2:26:57
Sicario	Movie, R	1698095506321	2023-10-23 21:11:46	0:29:09 / 2:01:33
The Peripheral, "Pilot"	Episode	1698095645376	2023-10-23 21:14:05	0:24:22 / 1:12:47
Memory	Movie	1698096528354	2023-10-23 21:28:48	0:53:10 / 1:53:38

lastAccessed is the last time each title was opened, and watched_position is where playback stopped. Neither proves the title was watched straight through. Sicario, The Peripheral and Memory were opened within 17 minutes of each other, which looks like browsing between titles rather than viewing.

The cached Profiles response under files/responseCache/.../Profiles lists three Prime Video profiles: Sarfraz (ADULT, default, ID A2YI4Z0W6FQDFA), Saif (CHILD) and Kids (CHILD) (Figures 80 and 81). All playback rows belong to the adult profile. Anyone at the device could have used that profile.

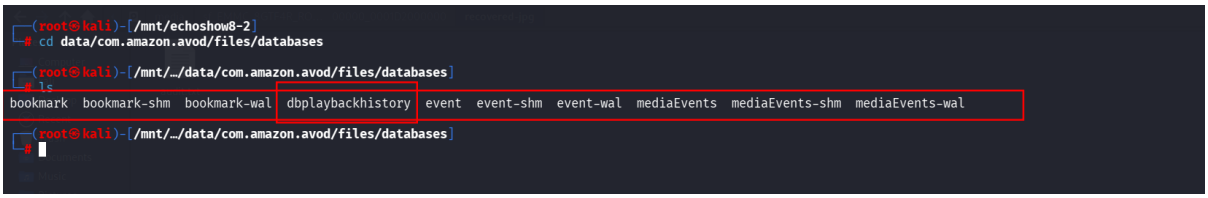


Figure 78: Prime Video database folder

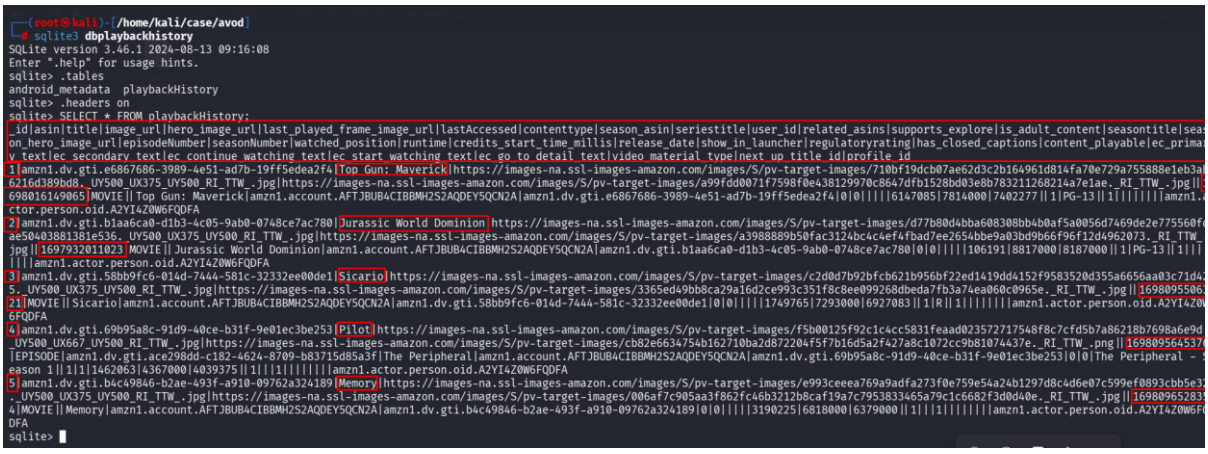


Figure 79: playbackHistory: five titles with lastAccessed values



Figure 80: Cached Prime Video profiles: Sarfraz (adult, default)



Figure 81: Cached Prime Video profiles: Saif and Kids (child)

Corroboration. PhotoRec recovered a fragment, f1271632.elf, from userdata unallocated space. It contains TTML subtitle lines in "AmazonDefaultStyle", such as "Only man to shoot down three enemy planes in the last 40 years" and "Thirty seconds to Tomahawk impact on enemy airstrip" (Section 4.7). These lines fit Top Gun: Maverick and support the playback record, but the fragment has no timestamp of its own.

Photo Booth videos. The two MP4 uploads are listed in Section 4.4. A foremost MP4 carve over the whole image recovered 0 files (Figures 82 and 83). The originals may have been removed after upload and their blocks reused, the files may be fragmented, or their header may not match foremost's MP4 rule. Without a recovered file there is no duration or content to report, only the size and upload time.

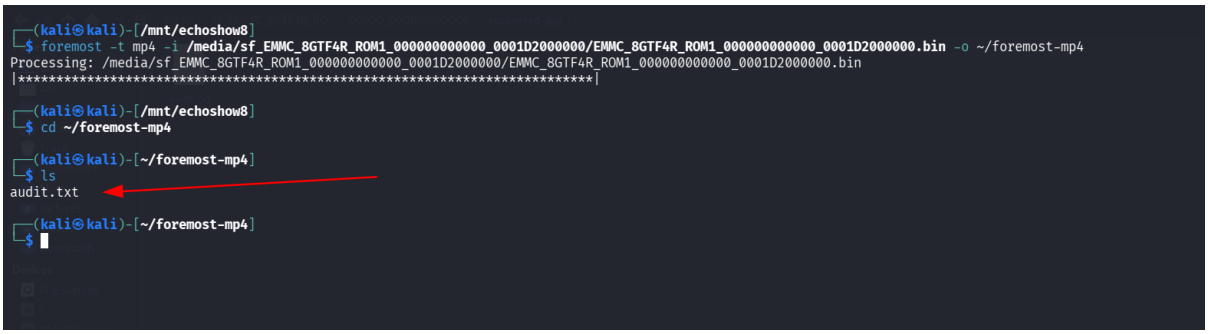


Figure 82: foremost MP4 carve of the full image

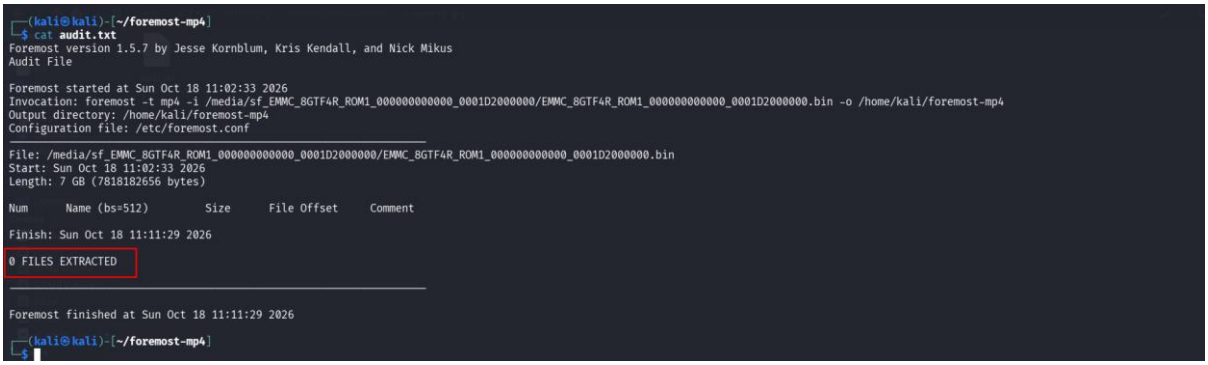


Figure 83: foremost MP4 audit: 0 files extracted

4.6 Audio and Music

Source: data/com.amazon.paladin/files/alexa-anchor/file.0000000000000000237 (228,663 bytes) and message-timestamps.0000000000000000237. The first file is one line of JSON that stops before its closing brackets, so jq would not parse it. On a copy in /home/kali/case/paladin I located the unclosed brackets, confirmed the last record was complete, and appended]} to make file237_fixed.json. It contains 248 events (Figures 84 to 90).

Every event uses schema alexa_anchor.DeviceImpression.3. Each one is a home-screen card that was shown and then closed (IMPRESSION_END, duration in ms, dismissalReason AUTO). They are display records, not playback logs. The message-timestamps file runs from 2023-10-23T18:28:05.498−0400 to 19:21:57.084−0400, which is 22:28:05 to 23:21:57 UTC (Figures 96 and 97).

Time (−0400)	UTC	Card	Content
18:28:05.498	22:28:05	PhotoCard-AmazonPhotos	First record; deviceState camera ON, privacyMode OFF, cameraShutter OFF
18:28:44.594	22:28:44	RecentlyPlayedAudioContentProvider	Amazon Music SONG "money heist theme song (Bella Ciao - lofi edit)"
18:41:55.528	22:41:55	RecentlyPlayedAudioContentProvider	Amazon Music SONG "Besharam Rang (From "Pathaan")"
18:59:32.561	22:59:32	PhotoCard-AmazonPhotos	Photo frame card
19:17:13.889	23:17:13	RecentlyPlayedAudioContentProvider	Amazon Music SONG "Centuries"; personId UNDEF
19:18:47.391	23:18:47	RecentlyPlayedAudioContentProvider	Amazon Music SONG "Mix"

A "recently played" card means the song had been played before the card appeared, so each card time is an upper limit for the play time, not the play time itself. Early records carry personId amzn1.actor.person.did.AMFXG2V..., while the later records show "UNDEF". The device stopped linking

impressions to a recognised person during the evening, and the log does not say why. No audio files were recovered, and I did not attempt voice comparison or transcription.

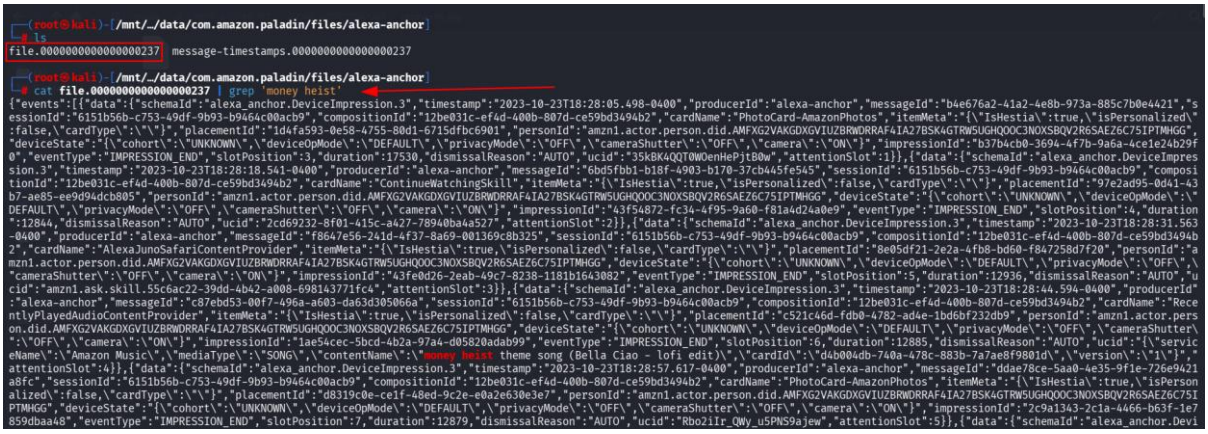


Figure 84: alexa-anchor folder and raw grep for "money heist"

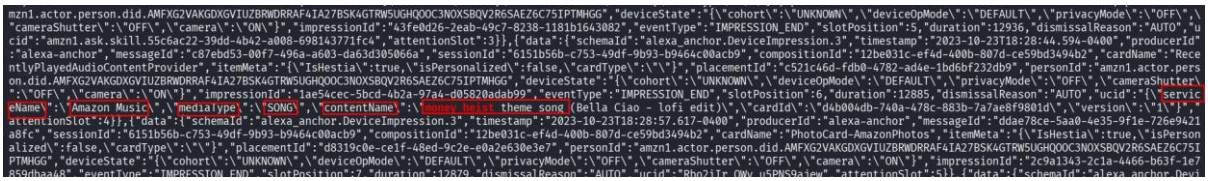


Figure 85: Raw record: Amazon Music, SONG, money heist theme song

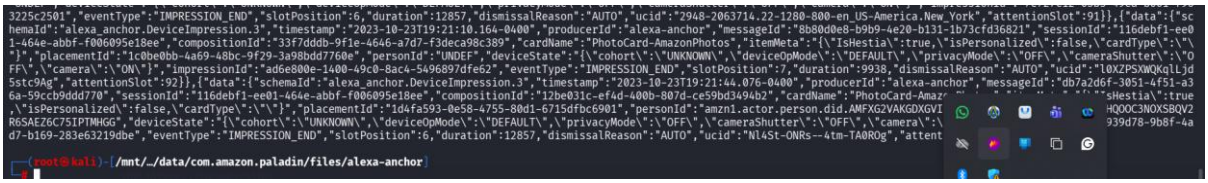


Figure 86: End of the raw file: last record closes with "}}", array never closed

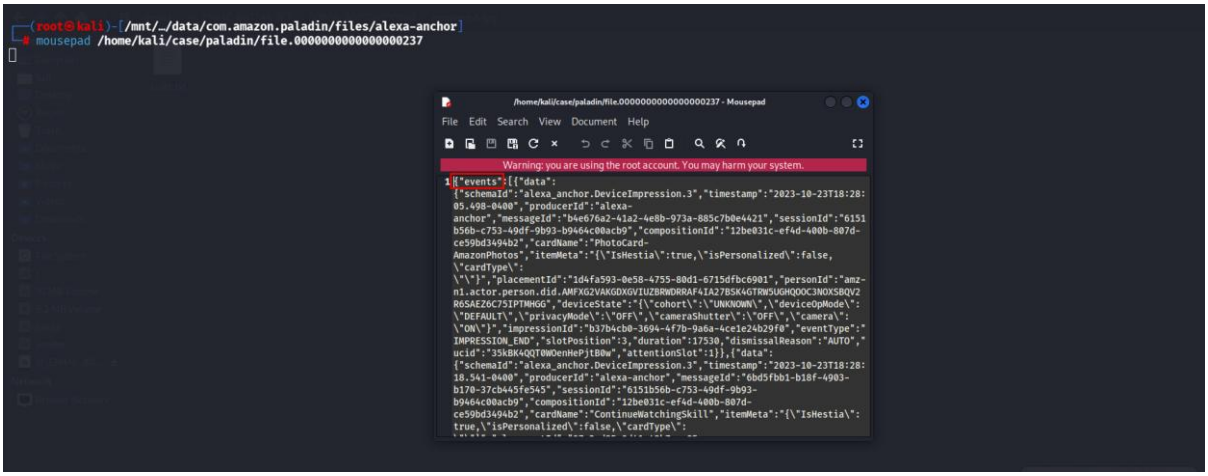


Figure 87: Raw file opened from the case copy in Mousepad



Figure 88: First parsed event from file237_fixed.json

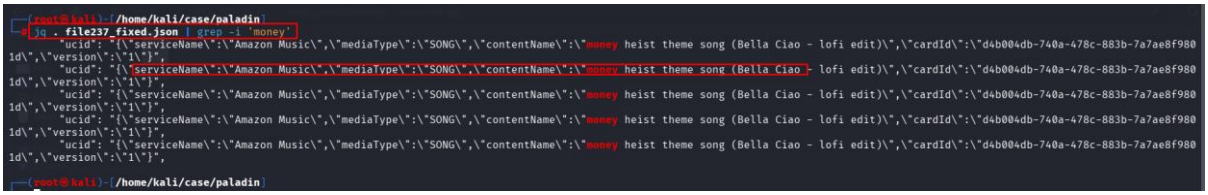


Figure 89: jq output piped to grep: five money heist matches

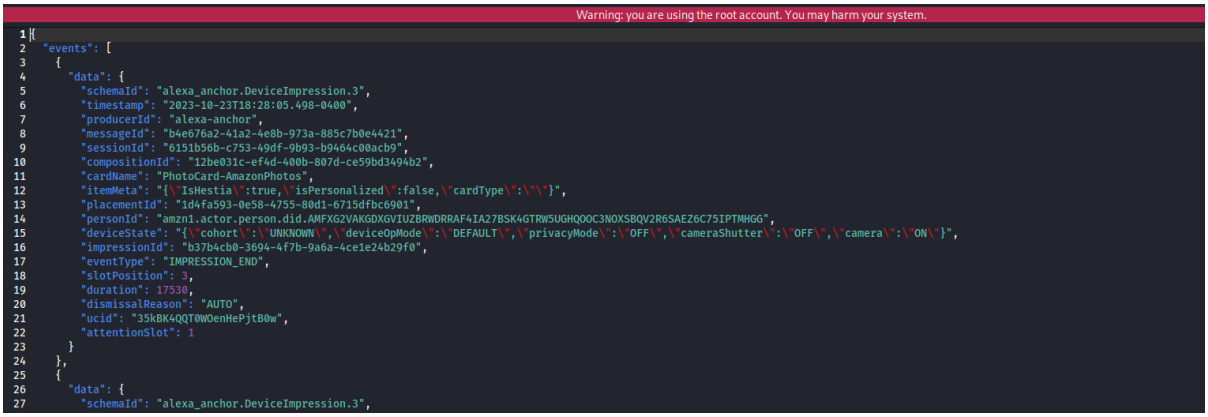


Figure 90: Repaired JSON viewed as formatted text

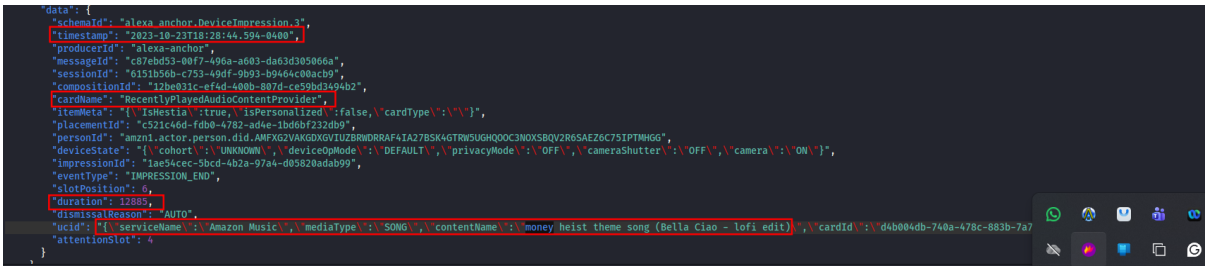


Figure 91: Money heist theme song card at 18:28:44 −0400

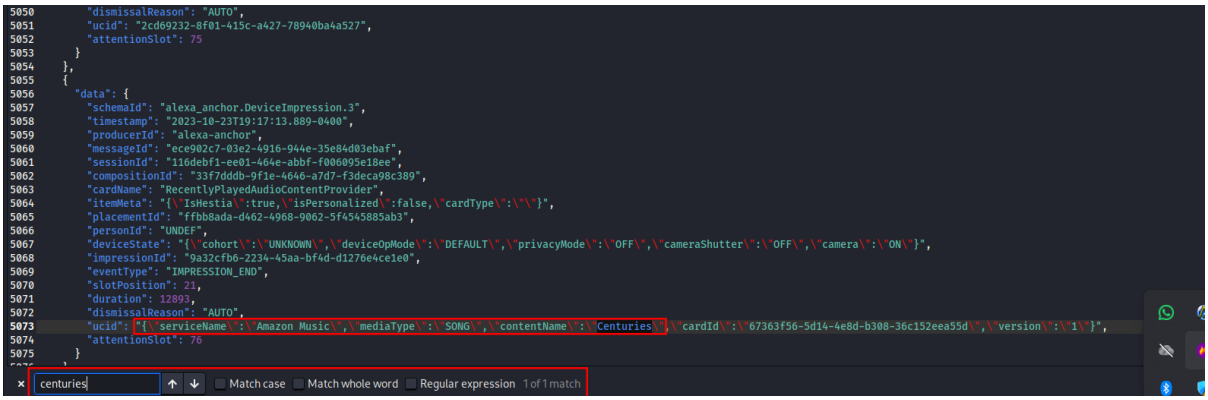


Figure 92: "Centuries" card at 19:17:13 −0400



Figure 93: "Mix" card at 19:18:47 −0400



Figure 94: "Besharam Rang" card at 18:41:55 −0400



Figure 95: Photo frame card (PhotoCard-AmazonPhotos)

```
┌──(root㉿kali)-[/mnt/…/data/com.amazon.paladin/files/alexa-anchor]
└─# ls
file.0000000000000000237  message-timestamps.0000000000000000237

┌──(root㉿kali)-[/mnt/…/data/com.amazon.paladin/files/alexa-anchor]
└─# cat message-timestamps.0000000000000000237
2023-10-23T18:28:05.498-0400
2023-10-23T18:28:18.541-0400
2023-10-23T18:28:31.563-0400
2023-10-23T18:28:44.594-0400
2023-10-23T18:28:57.617-0400
2023-10-23T18:29:10.642-0400
2023-10-23T18:29:11.370-0400
2023-10-23T18:29:11.405-0400
2023-10-23T18:29:19.938-0400
2023-10-23T18:29:21.945-0400
2023-10-23T18:29:22.911-0400
2023-10-23T18:29:22.922-0400
2023-10-23T18:29:22.949-0400
2023-10-23T18:29:24.826-0400
2023-10-23T18:29:31.429-0400
2023-10-23T18:29:31.445-0400
2023-10-23T18:30:02.439-0400
2023-10-23T18:30:22.567-0400
2023-10-23T18:30:25.264-0400
2023-10-23T18:30:59.205-0400
2023-10-23T18:31:12.220-0400
2023-10-23T18:31:25.243-0400
2023-10-23T18:31:38.268-0400
2023-10-23T18:31:51.276-0400
2023-10-23T18:33:01.764-0400
2023-10-23T18:33:14.773-0400
2023-10-23T18:33:27.781-0400
2023-10-23T18:33:40.791-0400
```

Figure 96: message-timestamps file: first entries

```
2023-10-23T19:19:00.401-0400
2023-10-23T19:19:13.414-0400
2023-10-23T19:19:26.443-0400
2023-10-23T19:19:31.625-0400
2023-10-23T19:20:08.029-0400
2023-10-23T19:20:21.055-0400
2023-10-23T19:20:34.080-0400
2023-10-23T19:20:47.091-0400
2023-10-23T19:21:00.116-0400
2023-10-23T19:21:10.164-0400
2023-10-23T19:21:44.076-0400
2023-10-23T19:21:57.084-0400
```

Figure 97: message-timestamps file: last entry 19:21:57.084 −0400

4.7 Unallocated Space (PhotoRec)

I ran PhotoRec 7.2 on the userdata partition: ext2/3/4 filesystem, "Free" (unallocated space only), with only the ELF file type selected. The run that saved results used the .bin image directly with partition 23 (userdata) selected, and it wrote 91 files to /home/kali/case/photorec/recup.1 (Figures 98 to 109).

grep -r "money" matched 14 of the carved files. strings on f5777032.elf shows base64 blocks and JSON with signature and timestamp fields (1697908517 = 2023-10-21 17:15:17 UTC). f1271632.elf holds the Top Gun subtitle text described in Section 4.5. The contents are not executable code. PhotoRec cut these blocks at an ELF-like signature, so the .elf extension is misleading, and each file name is just the sector where the block was found. These are deleted fragments with no original name or timestamp.

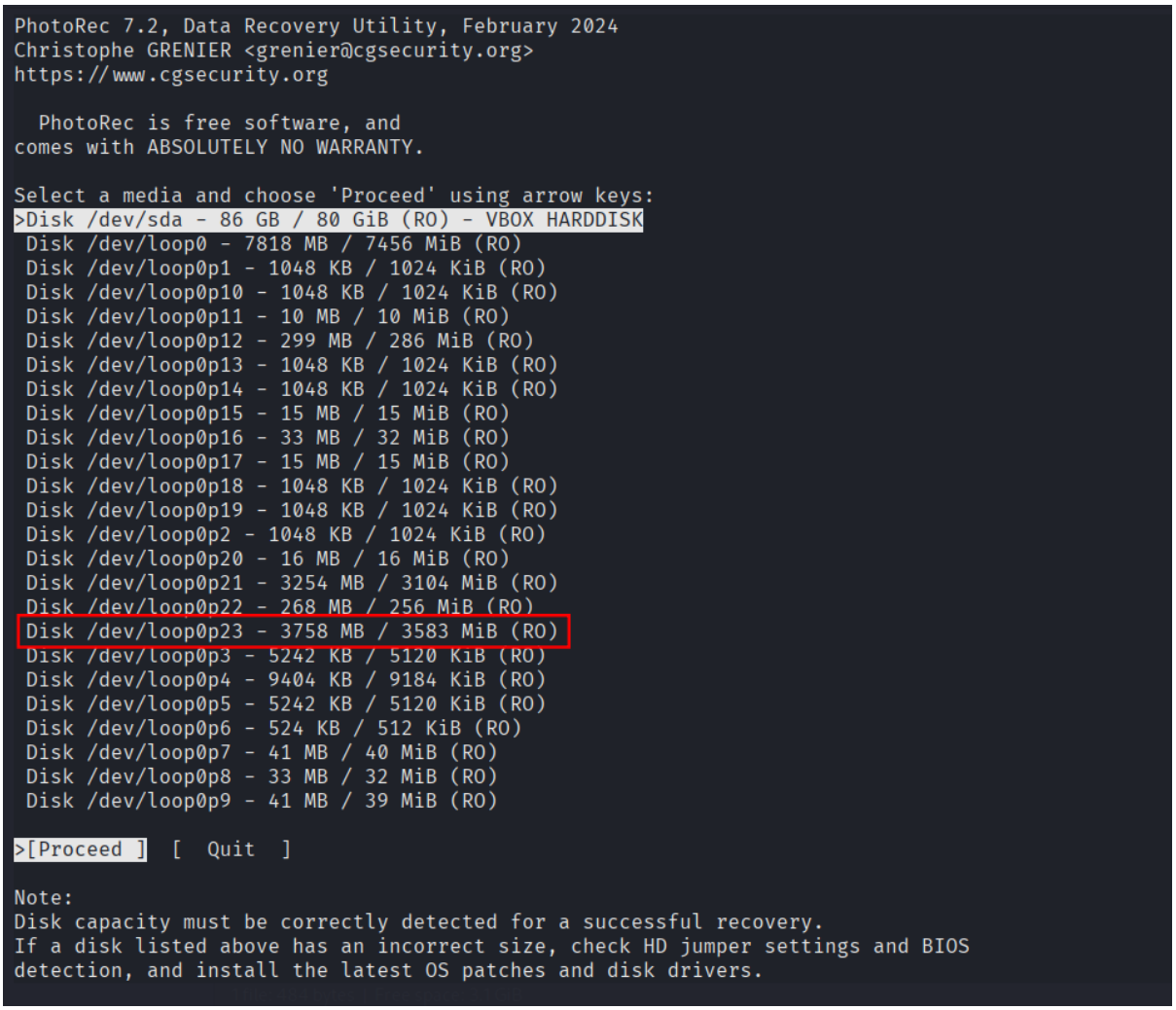


Figure 98: PhotoRec media list; /dev/loop0p23 read-only selected

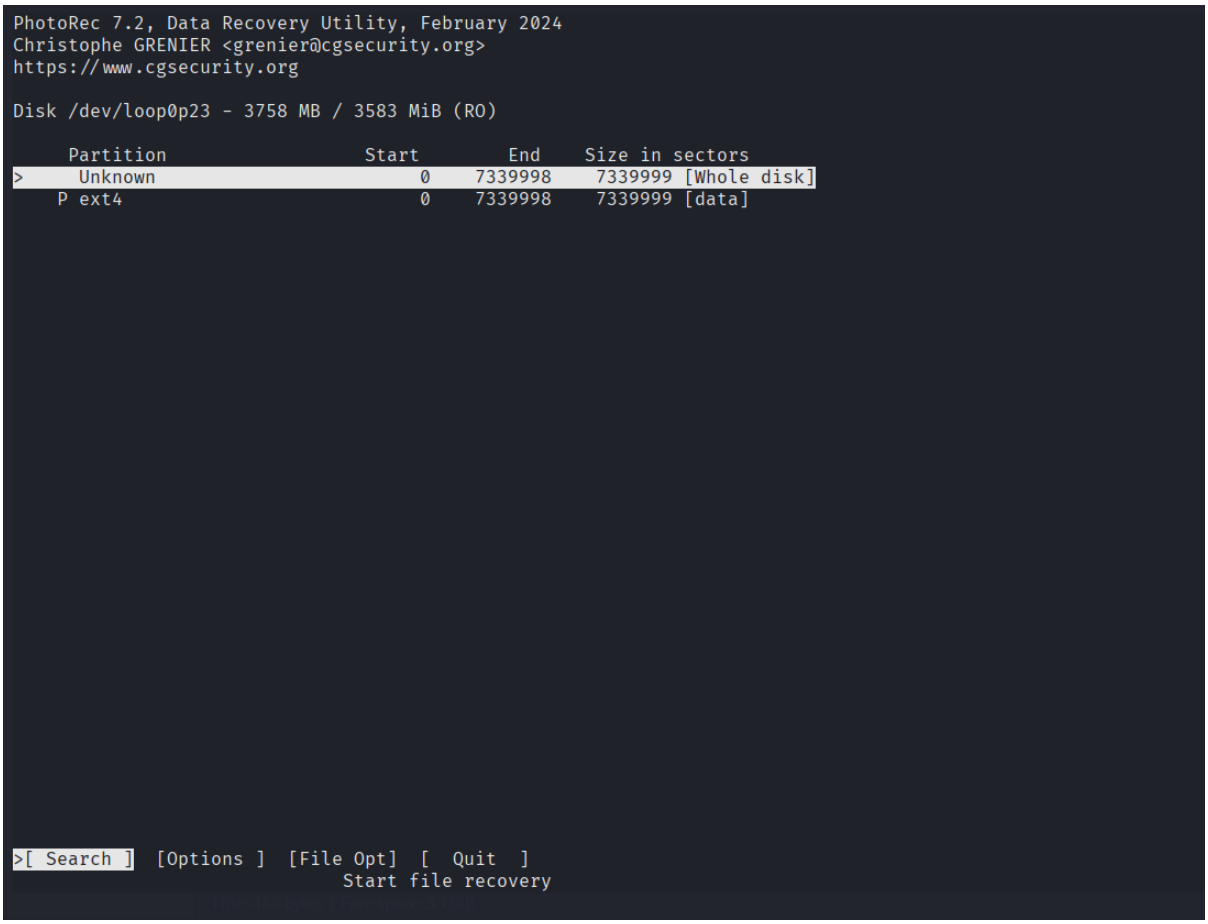


Figure 99: PhotoRec partition view of loop0p23

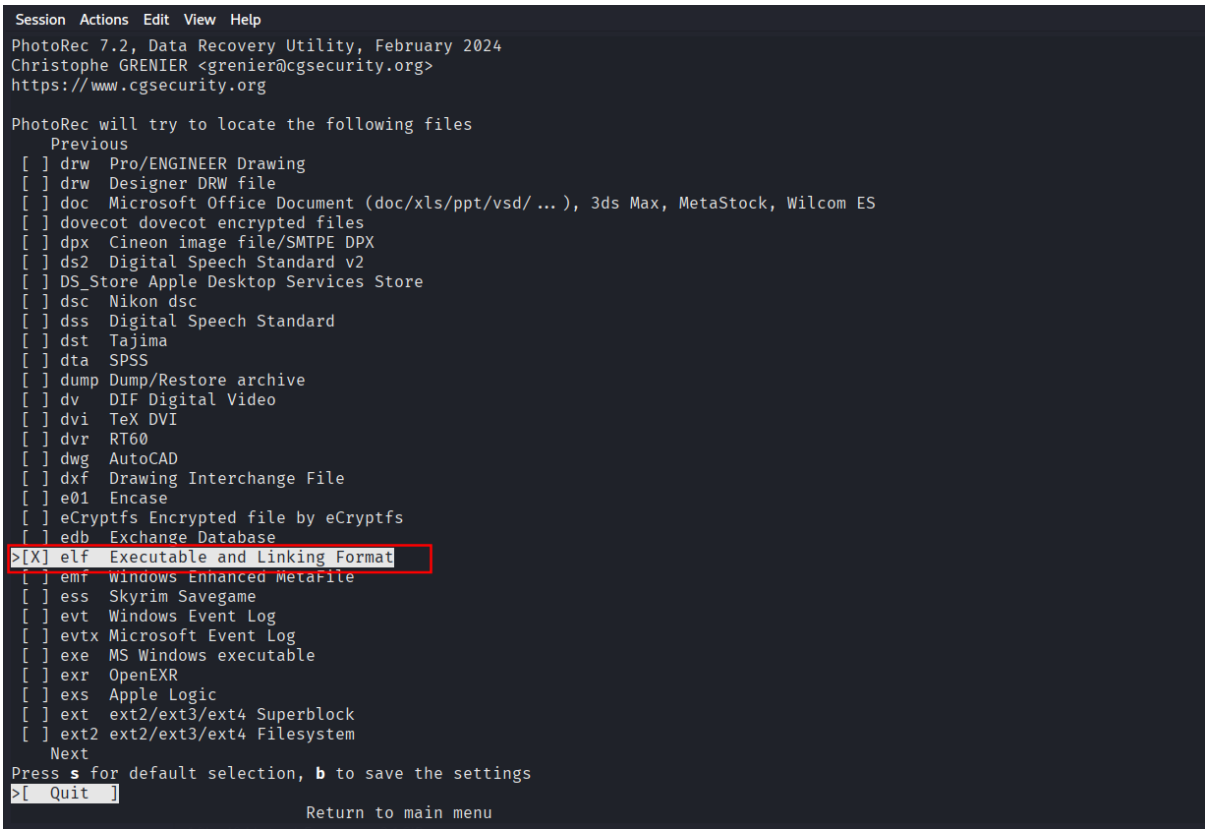


Figure 100: PhotoRec file options: ELF selected

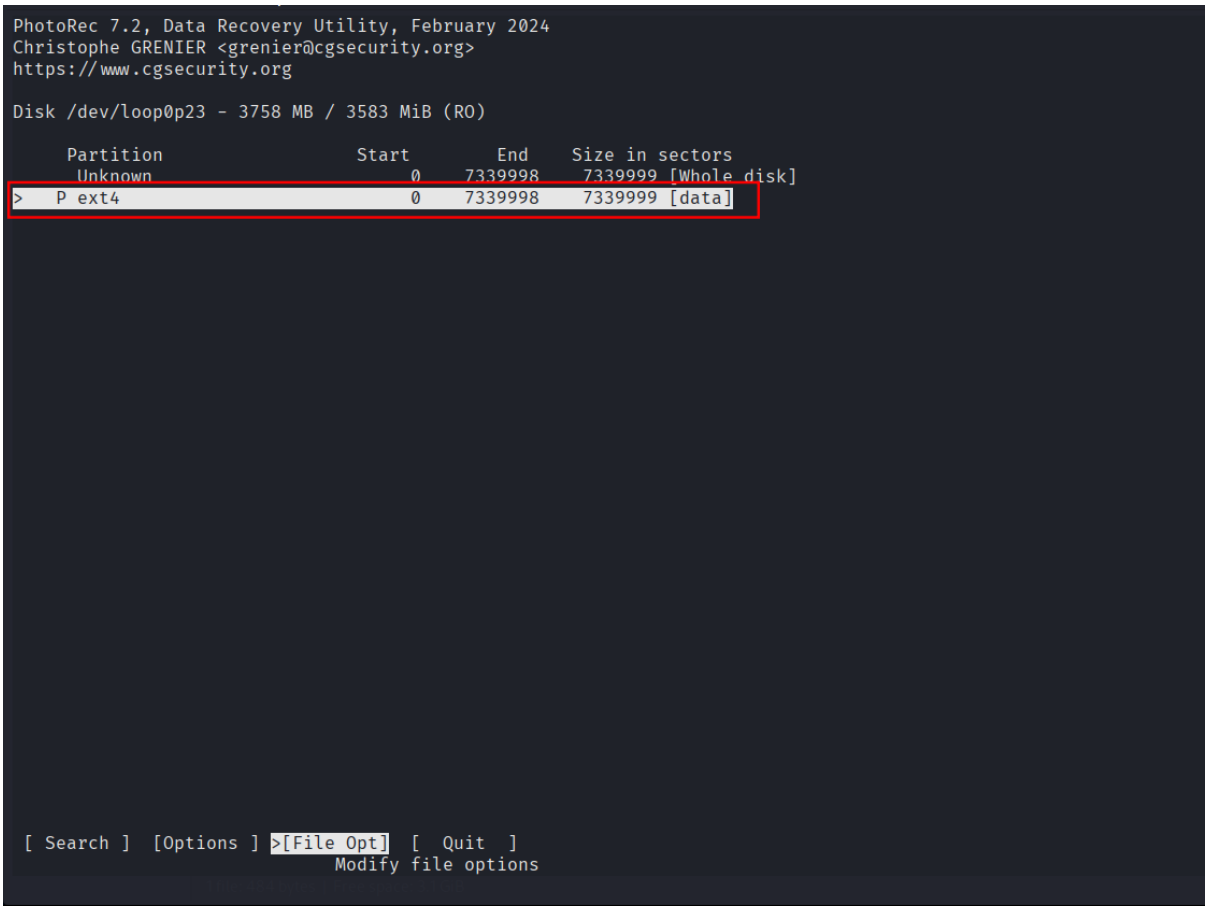


Figure 101: ext4 data partition selected, File Opt

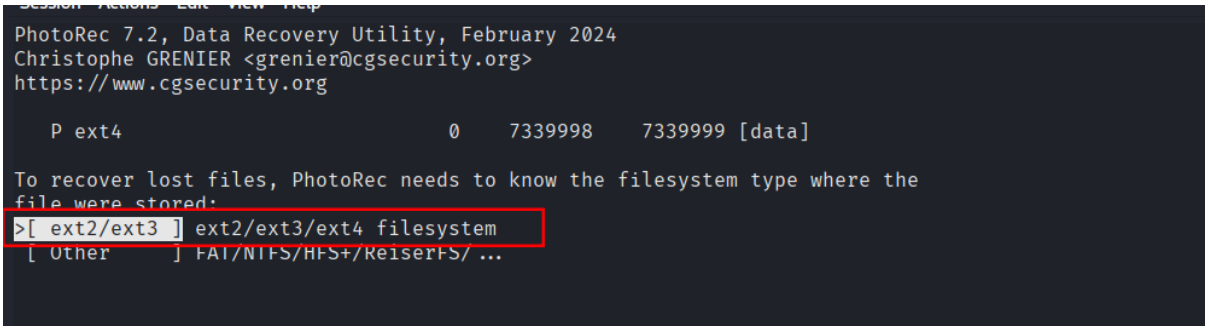


Figure 102: Filesystem type ext2/ext3/ext4

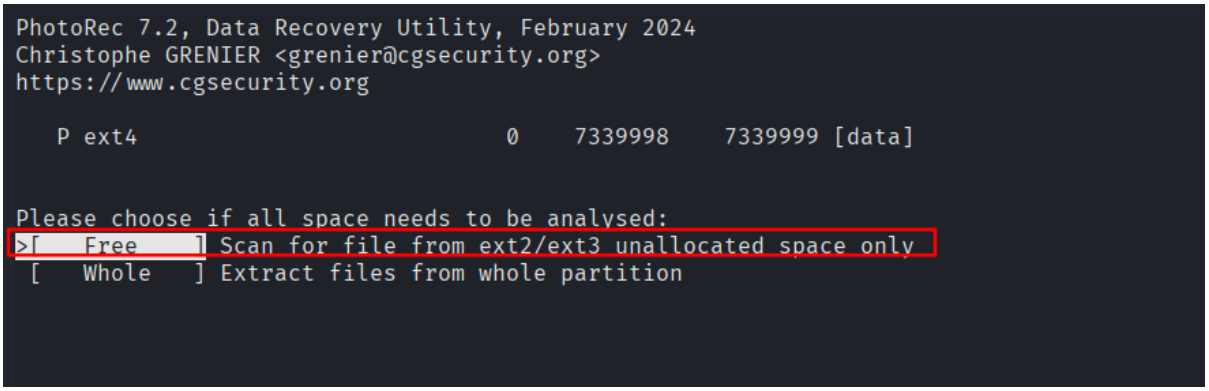


Figure 103: Free (unallocated space only) chosen

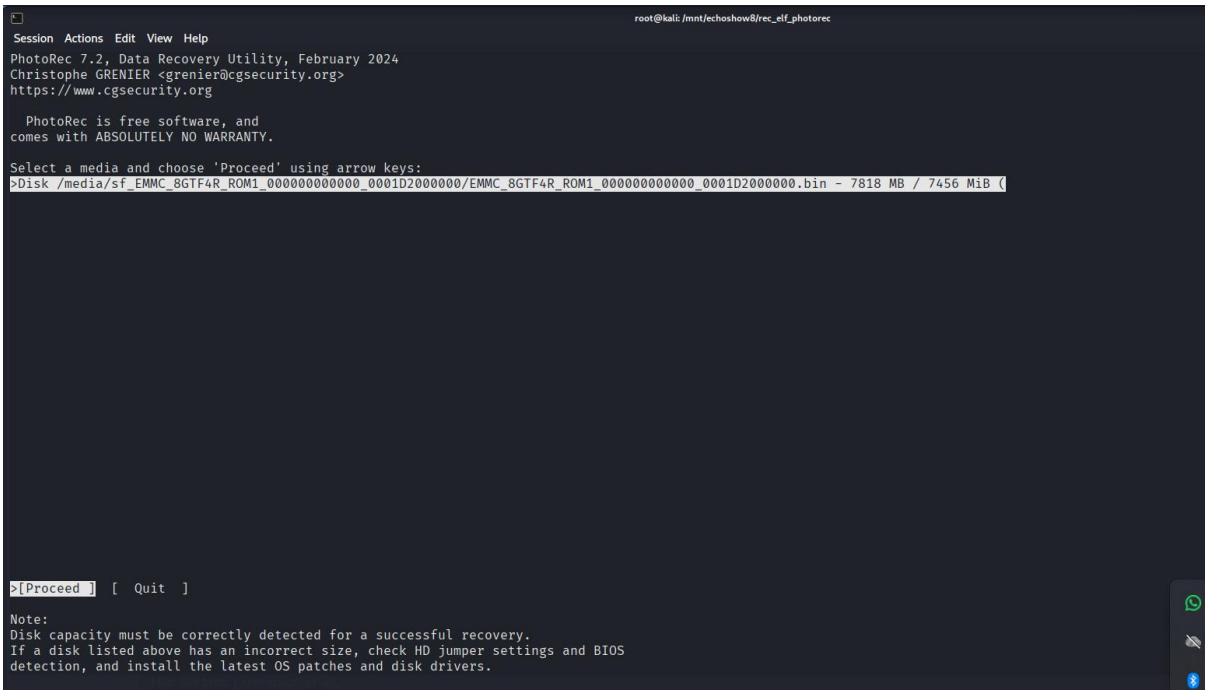


Figure 104: PhotoRec opened on the eMMC .bin image

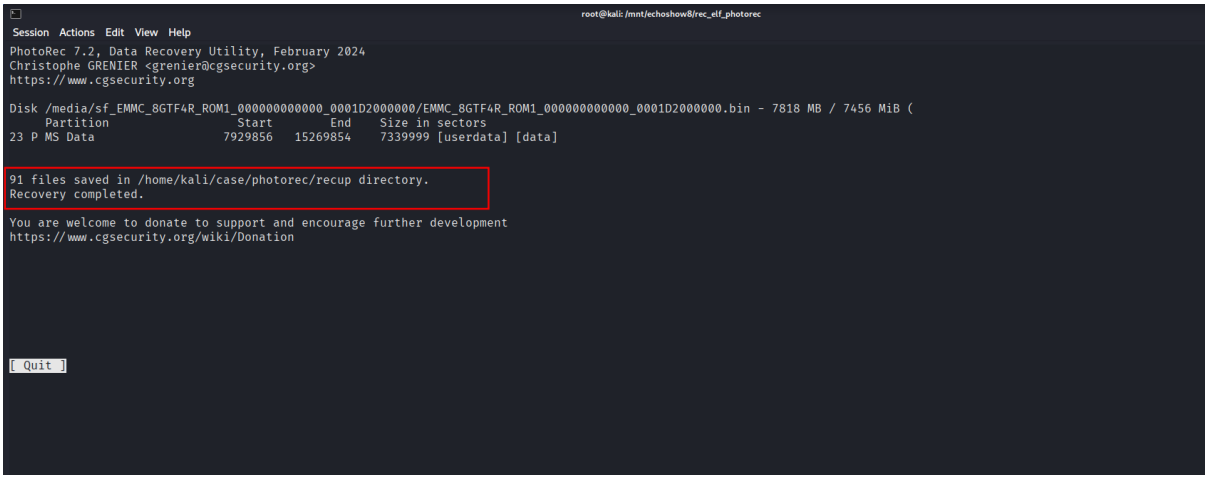


Figure 105: Recovery complete: 91 files saved from userdata (partition 23)

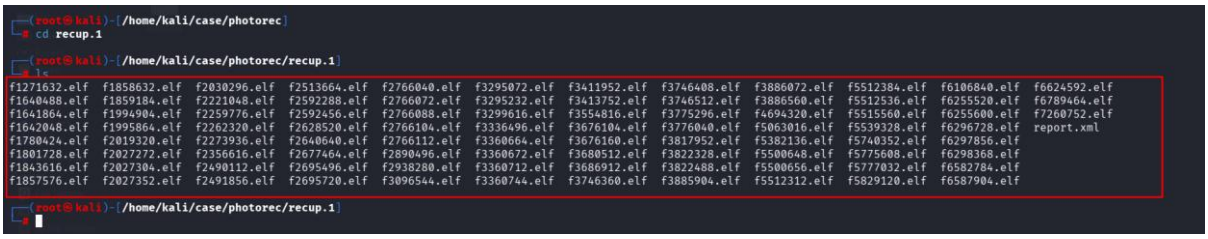


Figure 106: Carved files in recup.1

```
┌──(root㉿kali)-[/home/kali/case/photorec/recup.1]
└─# grep -r "money" .
grep: ./f6106840.elf: binary file matches
grep: ./f7260752.elf: binary file matches
grep: ./f2695720.elf: binary file matches
grep: ./f2221048.elf: binary file matches
grep: ./f3413752.elf: binary file matches
grep: ./f2766040.elf: binary file matches
grep: ./f3360744.elf: binary file matches
grep: ./f3746408.elf: binary file matches
grep: ./f5512536.elf: binary file matches
grep: ./f3886560.elf: binary file matches
grep: ./f5512384.elf: binary file matches
grep: ./f3686912.elf: binary file matches
grep: ./f1271632.elf: binary file matches
grep: ./f5777032.elf: binary file matches

┌──(root㉿kali)-[/home/kali/case/photorec/recup.1]
└─#
```

Figure 107: grep for "money" across carved files



Figure 108: strings on f5777032.elf



Figure 109: TTML subtitle text in f1271632.elf

5. Event Chronology

Window: 2023-10-21 22:00 UTC to 2023-10-23 23:30 UTC (about 49.5 hours). I chose this window because it holds the densest run of activity across independent sources: the last user login, the last web visits, the newest recent task, Prime Video playback, a Photo Booth capture and upload, and the whole alexa-anchor log. Local device time is EDT (UTC−4).

#	UTC	Source	Raw value	Event	Qualification
1	21 Oct 22:24:48	Web view History, urls id 52	13342400688326954	Alexa and Alexa Device FAQs page loaded	Page load only
2	21 Oct 23:02:39	Zordon upload_requests row 7 (filename)	2023-10-21_19-02-39 (EDT)	Photo Booth photo taken	Local-time filename; see #3
3	21 Oct 23:02:52	Zordon cloud_nodes created_date	1697929372480	Same photo registered for upload, DONE	Corroborates #2 (13 s later)
4	21 Oct 23:03:49	Zordon upload_requests row 8 (filename)	1697929429134	Photo Booth video saved	Epoch filename
5	21 Oct 23:04:24	Zordon cloud_nodes created_date	1697929464021	Video upload record created	Corroborates #4 (35 s later)
6	21 Oct 23:46:51	Prime Video playbackHistory	1697932011023	Jurassic World Dominion opened, stopped at 1 m 46 s	Last access only
7	22 Oct 00:29:05	Web view History, urls id 56	13342408145145602	Microsoft sign-in (sarfy53@hotmail.com) for Alexa calendar link	Page load; see #8
8	22 Oct 00:29:39	Web view History, urls ids 60 to 62	13342408179474433	"Account Linked" confirmation	Corroborates #7
9	22 Oct 01:10:29	Web view History, urls id 63	13342410629408781	YouTube search "kids videos"	User not identified
10	22 Oct 23:09:09	Prime Video playbackHistory	1698016149065	Top Gun: Maverick opened, stopped at 1 h 42 m	Supported by carved subtitles (no time)
11	22 Oct 23:16:36	system/users/0.xml	1698016596033	User 0 (Sarfraz) lastLoggedIn	Android user session, not a person
12	22 Oct 23:50:12	History + keyword_search_terms	13342492212049982	Google search "close google"	Two tables agree
13	22 Oct 23:50:50	History + keyword_search_terms	13342492250476128	Google search "search chicken nuggets on google"	Two tables agree
14	23 Oct 00:43:53	History visits id 126	13342495433605931	Last web view visit	End of web record
15	23 Oct 00:56:12	recent_tasks/617_task.xml + snapshot 617.jpg	1698022572750	YouTube search intent "kapil sharma show"; snapshot shows the clip	Matches urls id 50 (20 Oct)
16	23 Oct 21:11:46	Prime Video playbackHistory	1698095506321	Sicario opened	Last access only
17	23 Oct 21:14:05	Prime Video playbackHistory	1698095645376	The Peripheral, "Pilot" opened	Last access only
18	23 Oct 21:28:48	Prime Video playbackHistory	1698096528354	Memory opened, stopped at 53 m	Last access only
19	23 Oct 22:28:05	alexa-anchor + message-timestamps	2023-10-23T18:28:05.498-0400	First home-screen impression; camera ON	Both files agree
20	23 Oct 22:28:44	alexa-anchor	2023-10-23T18:28:44.594-0400	Recently played: money heist theme song	Played before this time
21	23 Oct 23:17:13	alexa-anchor	2023-10-23T19:17:13.889-0400	Recently played: Centuries; personId UNDEF	Played before this time

#	UTC	Source	Raw value	Event	Qualification
22	23 Oct 23:21:57	message-timestamps	2023-10-23T19:21:57.084-0400	Last anchor record	End of log

Corroborated relationships: photo capture and upload (#2 and #3); video save and upload (#4 and #5); Microsoft sign-in and account link (#7 and #8); Google searches in two tables (#12 and #13); the Kapil Sharma task, its snapshot and the earlier web search (#15); Top Gun playback and the carved subtitle fragment (#10); and the anchor log against its timestamp file (#19 to #22).

6. Interpretation and Limits

Finding 1: Device and configuration. The image comes from an Amazon AEOAT ("athena") device running Fire OS 7.5.6.0 (Android 9, build PS7560.3677N). It carries serial G6G1GG10141403SD, friendly name "Amazing Assistant" and Bluetooth name "Echo Show 8 (2nd Gen)-3SD". The serial and name each appear in two separate files. The "Echo Show 8 (2nd Gen)" label rests on the Bluetooth name alone. The image says nothing about the chip-off procedure beyond its file name and size, and it cannot show the state of a live device.

Finding 2: User and application activity. There is one Android user (0, "Sarfraz") and one Amazon account, re-added on 9 October 2023. These are account labels. They do not identify the person using the screen. YouTube searches for kids' content and the child Prime Video profiles suggest a child may also have used the device, but every playback row sits under the adult profile, so the records cannot separate the users.

Finding 3: Network, web and media. The device saved two Wi-Fi networks, paired with a MacBook and a speaker, linked a Microsoft account for calendar access, and took and uploaded six photos and two videos. Wi-Fi and Bluetooth entries are configuration and do not log connection times. Web visits are page loads. Anchor records are card displays, not plays. Carved files have no original names or times. No phone communication records were found.

Finding 4: Chronology. Within the window, the device was in use on the evenings of 21, 22 and 23 October local time. Photo, web, Prime Video and home-screen records follow the same −4 hour offset. I did not check the device clock against an outside source. The 2 October setup entries appear in UTC, while later ones use EDT, so any time near setup should be read with that change in mind. The same applies to the nextSerialNumber of 10 against a single user, which I could not explain from the image.

7. Appendices

Appendix A: Findings Log

ID	Finding	Source	Raw value	Converted (UTC)	Qualification
F01	Image hashes	eMMC .bin	MD5 b58fa049…; SHA-256 8258ec28…	n/a	Computed by examiner; no supplied checksum
F02	Build and OS	p21 /system/build.prop	ro.build.date.utc=1695264110	2023-09-21 02:41:50	Matches 0.xml fingerprint
F03	Serial and friendly name	p23 DNSVR_STORE.xml; device_capabilities file	G6G1GG10141403SD; Amazing Assistant	n/a	Two sources agree
F04	User 0 last login	p23 system/users/0.xml	1698016596033	2023-10-22 23:16:36	Account label only
F05	Account re-added	p23 accounts_de.db accounts and debug_table	1696883981793; 16:39:41 local	2023-10-09 20:39:41	Debug entry 4 h behind: EDT
F06	Address response	p23 otter.oobe volley cache	Date: Mon, 09 Oct 2023 20:40:08 GMT	2023-10-09 20:40:08	Shipping address from Amazon, not GPS
F07	Paired MacBook	p23 bt_config.conf; Bt_Metrics; snapshot 623	f4:5c:89:c7:54:f0	n/a	Raw MAC histogram agrees
F08	Microsoft account link	p23 cloud9 History urls 56, 60 to 62	13342408145145602 / 13342408179474433	2023-10-22 00:29:05 / 00:29:39	Cloud content out of scope
F09	Google searches	p23 History urls 69, 70; keyword_search_terms	13342492250476128	2023-10-22 23:50:50	Two tables agree
F10	Kapil Sharma task	p23 recent_tasks/617_task.xml; snapshots/617.jpg	1698022572750	2023-10-23 00:56:12	Matches web search on 20 Oct
F11	Photo Booth uploads	p23 Zordon mixtape.db	1697929372480	2023-10-21 23:02:52	Filenames in EDT
F12	Prime Video playback	p23 avod dbplaybackhistory	1698016149065	2023-10-22 23:09:09	Carved subtitles support
F13	Music card	p23 paladin alexa-anchor file 237	2023-10-23T18:28:44.594-0400	2023-10-23 22:28:44	Impression, not play
F14	Phone records	Users, usage stats, Bluetooth profiles	None found	n/a	Not established

Appendix B: Commands and Queries

```
md5sum EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin
sha256sum EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin
sudo losetup --partscan --find --show --read-only EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin
lsblk /dev/loop0
mmls EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin
sudo mount /dev/loop0p21 /mnt/echoshow8
sudo mount -o ro,noload /dev/loop0p23 /mnt/echoshow8-2
dd if=EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin of=userdata_partition.bin bs=512 skip=7929856 count=7339999
bulk_extractor -o ~/bulkext_raw /media/sf_EMMC_.../EMMC_8GTF4R_ROM1_000000000000_0001D2000000.bin
sqlite3 map_data_storage_v2.db "SELECT * FROM accounts; SELECT * FROM encryption_data;"
sqlite3 accounts_de.db "SELECT * FROM accounts; SELECT * FROM meta; SELECT * FROM debug_table;"
sqlite3 History "SELECT * FROM visits; SELECT * FROM urls; SELECT * FROM keyword_search_terms;"
sqlite3 dbplaybackhistory "SELECT * FROM playbackHistory;"
```

sqlite3 amzn1.account.AFTJBUB4CIBBMH2S2AQDEY5QCN2A.mixtape.db "SELECT * FROM upload_requests; SELECT * FROM cloud_nodes;"
scalpel -o recovered-xml userdata_partition.bin
foremost -t jpg -i <image>.bin -o ~/foremost-jpg
foremost -t mp4 -i <image>.bin -o ~/foremost-mp4
exiftool 01271252.jpg
{ cat file.0000000000000000237; printf ']}'; } > file237_fixed.json
jq '.events | length' file237_fixed.json
jq . file237_fixed.json | grep -i money
photorec (loop0p23, ext2/ext3, Free, ELF only, destination /home/kali/case/photorec)
strings f1271632.elf | grep enemy
date -u -d @<epoch_seconds>

Appendix C: Manual Validation

Tool result	Manual check	Outcome
jq parse of file237_fixed.json: money heist card at 18:28:44.594−0400 (Figure 91)	grep on the raw, unrepaired file shows the same record and fields (Figures 84 and 85)	Matched. The repair only appended "]}" and did not change any record.
jq .events[0].timestamp = 2023-10-23T18:28:05.498-0400 (Figure 88)	First line of message-timestamps.0000000000000000237 (Figure 96)	Matched to the millisecond.
bulk_extractor ether histogram: top MAC F4:5C:89:C7:54:F0 (Figure 44)	bt_config.conf and Bt_Metrics_Device_Connected.xml read directly (Figures 40 to 43)	Matched. The same address is the paired MacBook.
date -u conversion of accounts_de time 1696883981793	debug_table text entry 2023-10-09 16:39:41	Differed by exactly 4 h, explained by the debug_table using local EDT.

Appendix D: Academic Integrity Declaration

I confirm that this report reflects my own examination of the authorised course evidence and that I have identified all sources, tools, and assistance used.

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Registration Number: C11/26/DFIT/17302

Date: 18 October 2026